ROYAL CANADIAN NAVY
QUALIFICATION STANDARD
AND PLAN (QSP)

Marine Technician (MAR TECH) (00379)

RQ MARTECH PO2 (ALUW)

MITE Course IDs:
RQ MARTECH PO2 – 121982
DRMIS Course Mod 3 – DRMIS Maintenance Manager – 119098
Naval Technical Communication – Mod C – 118130

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Fleet School Pacific NFS (P). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0	18/05/2023	CNPTG

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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training and Personnel Group (CNPTG) and in accordance with the following references:

a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-002 Non-Commissioned Member General Specification dated 09 Feb 11;

c. A-PD-055-002/PP-002 Marine Technician Job Based Occupational Specification dated 04 April 17; and

d. NAVORD 4500-0 – Structured Delegation, and NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs), Enabling Objectives (EOs), and Lesson Specifications in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA, who will forward to NPTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS(P)	All	119098	ALUW
NFS(A)	All	119098	ALUW

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the MAR TECH Supervisor. This QSP is designed to assist training establishments in developing, conducting, evaluating, and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan, Annex B.

Allocation of ATT	Minutes
(Sum of all EOs/POs)	6510
Enabling Checks (ECs) (in-house)	2290
Performance Checks (PCs) (in-house)	1920
Actual Training Time (ATT) Sub-total	10720 (26.8 days)
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	120
Training Administrative Time (TAT)	120
Physical Fitness (PT) +50/PT Day TTT	1221.3
Sub-total	1461.3 (3.65 days)
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)	122201.3
For Scheduling duration total mins/400 = # days	**30.45 days**

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Snr Instructor	PO1 or above	Mech/Elec Manager (Legacy PO1)	1	All
Instructor	PO2 or above	Mech Supervisor (Legacy ER/Hull 6A)	1	All
Instructor	PO2 or above	Elec Supervisor (Legacy ETech 6A)	1	All

4. This table should be read in conjunction with the recommended course timetable in Annex G – Sequencing and Timings.

Critical Training Resources	No of Days
BDCS Trainer	1.25
SME lecturer (FTA)	0.25
SME lecturer (SEMS)	0.25
SME lecturer (Ship's Support Officer)	0.50
SME lecturer (FMF)	0.50
SME lecturer (Ship's CO)	.0375
Guest Lecturer - PCC	0.25
Guest Lecturer – Transition Cell Unit	0.25

COURSE CAPACITY

5. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 – Quantity Control Process for the NTS.

a. Maximum – 12;

b. Minimum – 3; and

c. Optimum – 10 (maximum number of seats in the trainer).

6. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

7. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System (NTS); in MITE by the NFS or Training Site (TS) conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends, and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

8. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. Primary Leadership Qualification (PLQ) (AINR) or equivalent;

b. An Engineering Watchkeeper Qualification from one of the following:

i. USQ EWK HFX (ALMJ);

ii. USQ EWK AOPV (NQUAL Pending);

iii. USQ EWK KIN (ALVB);

iv. USQ EWK VIC (NQUAL Pending);

v. USQ EWK ORCA (ALVC);

c. USQ – S1 Mech or Elec (ALUS or ALUT);

d. The rank of Master Sailor (MS);

e. DRMIS Course – Module 2 – 119097; and

f. General Safety Supervisor Course – 303409.

9. Qualification. Successful completion of this course constitutes the primary means of attaining the MAR TECH Supervisor (National Qualification(s) – ALUW). Additionally, the following MITE Course IDs will be granted:

a. DRMIS Course – Module 3 – 119098; and

b. Naval Technical Communication – Module C – 118130.

COURSE REPORTING

10. Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed DAOD 5031-9 – Course Reporting and Certificates and NPTORD 4500-51 – Course Reports and Certification.

EVALUATION

11. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 – Evaluation and Validation Techniques.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted in accordance with NAVORD 4500-27 – PLAR, DAOD 5031-1 – Canadian Forces Military Equivalencies Program and CFITES Vol 12 – Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. Joining instructions to candidates to include requirement to report on the first day with an approved topic for the technical writing component. Each candidate is to have suitable references identified, and preliminary research completed prior to reporting for duty. Approving authority for topics is NFS(A&P) MSE Div CPO.

CHAPTER 2: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF NFS (P)

GENERAL

14. This is a pass/fail course. Trainees must successfully complete all POs to pass this course. Performance Checks (PCs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification.

15. Trainees will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K (Assessment Plan) for details.

16. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in the convening of a Training Review Board (TRB).

ENABLING AND PERFORMANCE CHECKS

17. Enabling Checks (ECs). ECs are formative assessments that help the trainee, and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

18. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES Vol 6. with the unsatisfactory progress documented in their Student File.

19. Critical Enabling Checks (CECs) or Performance Checks (PCs). PCs / CECs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO or EO and hence can do the job. Supplemental PCs / CECs are administered upon failure of a PC / CEC and must be in the same format and have the same standard as the original PC / CEC.

20. PCs / CECs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO / EO.

21. All PC / CEC failures and supplemental failures are to be handled IAW NPTORD 4500-54. Supplemental PCs / CECs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.

RESPONSIBILITY FOR TESTING

22. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation of Learners.

23. NFS staff provides quality control by administering and scoring all PCs / CECs. Any PC not administered or invigilated by standards staff will be administered by a non-involved instructor designated by the senior instructor. ECs are the responsibility of the Instructional Staff.

PROGRESS MONITORING

24. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained. Formative assessment may include:

a. ECs;
b. Homework;
c. Assignments;
d. Quizzes;
e. Practice periods; and
f. Any other method or tool used to develop the trainee.

25. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 – Conduct of Instruction Programmes.

26. Continuous monitoring of candidate progress is required to provide the following:

a. Early warning of trainee difficulties;
b. Feedback on the effectiveness of training;
c. Information for any required TRBs; and
d. Information to assist in completion of the CF 377 Course Report.

27. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

28. Instructional staff shall maintain individual student files, including the results of all:

a. Student performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);
b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and
c. PCs / CECs.

TRAINING REVIEW BOARD

29. TRBs shall be convened and conducted IAW with NPTORD 4500-54 – Training Review Board Process.

30. In the event of prolonged trainee absence, Training Establishment Staff shall determine, on a case-by-case basis, the necessity to convene a TRB.

COURSE REPORTS

31. Course reports (CF 377) shall be prepared and distributed IAW DAOD, 5031-9 – Course Reporting and Certificates and amplified in NPTORD 4500-51 – Course Reporting and Certificates.

32. Trainees at the minimum will be graded pass or fail. Where necessary for Rank Qualifications, the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

33. The Course Report narrative comments could include:

a. Ability to work with others;
b. Ability to work under pressure;
c. Written and verbal communication;
d. Organizational skills; and
e. Leadership.

34. Training limitations and / or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF NFS (P)

PO 001 **PO 001 Draft technical service paper**

This PO will be captured by the Naval Technical Communication – Module C. MITE Code 118130 will be awarded on successful completion.

All timings, teaching points and assessments will be conducted IAW with Naval Technical Communication Mod C QSP.

PO 002 **PO 002 Act as Section Base I/C**

35. Performance. Act as Section Base I/C.

36. Conditions.

    a. Given:

        (1) Damage Control HQ;

        (2) Damage Control Section base;

        (3) Damage Control Organization;

        (4) Battle Damage Control System (BDCS);

        (5) References;

        (6) Supervision; and

        (7) Assistance.

    b. Denied: Nil.

    c. Environmental: Day or night, at sea or alongside, onboard ship or ashore.

37. Standard. The MAR TECH supervisor will act as section base team I/C (SBT IC) in support of the Damage Control organization, IAW refs, by:

    a. Making stability recommendations to HQ1;

    b. Coordinate section base team training by:

        (1) Training section base teams;

        (2) Providing opportunities for fire/flood training; and

        (3) Scheduling development opportunities and drills for junior personnel.

    c. Briefing team members on DC considerations; and

    d. Coordinating a section base team in actions and emergencies (IT0150).

38. References. A1, A2, A3, A4, A5, A6, A7, A8.

39. Task, Skill, Knowledge Numbers.

40. Training Limitations:

    a. Requires Damage control school; and

    b. Current Fit-test certification.

EO 002.01 Coordinate Section Base Team Training

41. Performance. Coordinate section base team training.

42. Conditions.

    a. Given:

        (1) References; and

        (2) Supervision.

    b. Denied: Assistance.

    c. Environmental: Classroom.

43. Standard. The trainee shall coordinate section base team training, IAW refs, by:

    a. Training section base teams to include:

        (1) Identifying the roles and responsibilities of each team member;

        (2) Describing shipboard damage control organization, to include the following:

            (a) Command;

            (b) HQ1;

            (c) Section Base teams;

            (d) Rapid Attack team (RAT);

            (e) Rapid Response Team;

            (f) Casualty Clearing teams;

            (g) CSE ERT; and

            (h) Manning pool.

        (3) Explaining the purpose of Damage Control Organization to include the following:

            (a) DC hierarchy;

            (b) Relationship between DC elements;

            (c) Command priorities for the following:

                (i) Action stations;

                (ii) Warfare priorities; and

                (iii) Damage Control events.

        (4) Explaining the purpose of key damage control personnel to include the following:

            (a) XO;

(b) EO;

(c) DCPO/Senior MAR TECH (Mech);

(d) CEng/Departmental Coord;

(e) SnrET/Senior MarTech (Elec);

(f) Fwd/Aft/3SBT I/C;

(g) Manning Pool I/C

(h) Snr ERT; and

(i) Snr P.A.

(5) Explaining requirements of damage control elements to include the following:

(a) Fire team;

(b) Flood team;

(c) Casualty power team;

(d) Bomb Threat response; and

(e) CBRN transit response.

(6) Explaining tasks performed by special teams including the following:

(a) RAT;

(b) ERT; and

(c) Electrical/mechanical repair team.

b. Scheduling section base team training.

c. Scheduling development opportunities and drills for junior personnel.

44. Time Required.

a. EO – 250 min; and

b. EC – 300 min.

45. Remarks. Nil.

46. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Describing the roles and responsibilities DC Organization			
1. DC hierarchy. 2. Purpose of Damage Control Organization.	A1, A2	**Method / Media:** Interactive lecture **Trg Environment:** Classroom **Activity Details:**	200 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
3. Purpose of key damage control personnel. 4. Requirements of damage control elements (including equipment). 5. Tasks performed by special teams.		PowerPoint lecture to include relevant information on the DC organization with instructor questions and feedback throughout (confirmation questions). **Trg Resources:** • Classroom; • IT assets (DWAN Workstation, projector); • SSOs; • Guest lecturer (DC SME from ST(P/A); someone that's lived through a real DC event); and • BOI results from HMCS PROTECTEUR (when released); photographs.	
LESSON OBJECTIVE: Developing a Section Base Team Training Plan			
1. Receiving and processing feedback. 2. Problem solving (determining the root cause of the deficiency). 3. Developing a plan to rectify (measurable, observable) ie. Competitions, practical training (Know Your Ship). 4. Scheduling and deconflicting. 5. Implementation.	A1, A2, A3	**Method / Media:** Guided discussion **Trg Environment:** Classroom **Activity Details:** Candidates to participate in group discussion, drawing upon their experiences as SBT members (critical thinking, reflection) **Trg Resources:** • SSOs; and • STG.	50 min

47. Test Details. EO 002.01 will be tested in EC 002.01 and PC 002.

EC 002.01 Coordinate Section Base Team Training

EC Time: 60 minutes for research; 20 minutes per candidate the next day for brief delivery and AAR.

ASSESSMENT ACTIVITY: Candidate will receive a brief from the PO1 about a deficiency (personnel performance, weak 2IC, plotting, DC team performance, etc.; each candidate will have a different deficiency within the same scenario) within the SB. The candidate will have to formulate a plan to correct this deficiency, oversee execution of training, and conduct an AAR on completion of directed training to reinforce the lessons identified. The candidate will brief the DCPO before the conduct of training to confirm the viability of the plan, and on completion of the AAR to report results.

Candidate will organize SBT Training to include but not limited to:
Table-Top, Lectures, Scenarios, Tours

Candidate will be assessed on the plan they formulate, and the briefs provided.

Candidate may create and use an Aide-Memoire (e.g. Duty Tech style index card)

ASSESSMENT RESOURCES:

- Scenario including a Section Base deficiency;
- Assessor to stand in as PO1 role player;
- DWAN workstation (for access to relevant DC resources);
- Section (other candidates); and
- FLEX (for exercise); watch rotation.

IMPACT OF FAILURE: If candidate is unsuccessful retrain and re-test.

MAR TECH RQ – PO2 – EC 002.01 Coordinate Section Base Team Training			
Candidate Name:	**Assessor Name:**	**Date:**	**Course session:**

INVIGILATOR NOTES:

1. Formative assessment activity.
2. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all critical checklist items & 7/10 overall.
3. IMPACT OF FAILURE: Remedial Training and Re-test.

ITEM	UNSAT	SAT	COMMENTS
1. Develop plan to correct SBT deficiency			
Identify root cause of deficiency (*Critical)			
Plan is feasible (*Critical)			
Brief plan to assessor (DCPO)			
2. Deliver an after-action report			
Introduction			
Review deficiency (*Critical)			
Review Intent			
Discussion of key events			
Self-assessment			
Review of Skills			
Summarize			

Briefing Note - ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.02 Brief team members on DC considerations

49. Performance: Brief team members on DC considerations.

50. Conditions.

    a. Given:

        (1) References;
        (2) DC Board; and
        (3) Supervision;

    b. Denied: Assistance; and

    c. Environmental: Classroom.

51. Standard. The trainee shall brief team members on DC considerations by:

    a. Reviewing the risk portfolio;

    b. Determining the optimal route to address the problem;

    c. Identifying potential hazards;

    d. Determining the requirement for additional equipment;

    e. Ensuring team safety; and

    f. Establishing comms with DC team.

52. Time Required:

    a. EO – 100 minutes to deliver training; and

    b. EC – 250 min for testing/quizzes.

53. Remarks. Nil.

54. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: ***At the end of this lesson, trainees will be able to brief section base 2I/C on DC considerations.***			
1. Review of Risk Portfolio. 2. Best way to attack a fire (same deck, below, above). Designating attack routes. 3. Types of hazards, to include:	A2, A3, A4, A5, A6, A7, A8	**Method / Media:** fishbowl exercise, EDI **Trg Environment:** Classroom **Activity Details:**	50 min EDI 50 min practice

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
a. Compressed Gas bottles, b. Electrical hazards; c. Fuel; and d. Smoke/fumes. 4. Additional equipment to be utilized to include: a. Extra hoses; b. Forced entry tool; c. Compartment keys; and d. TIC. 5. When to go on air. 6. Electrical power considerations. 7. Fitted systems. 8. Boundary and other locations and ensure status. 9. Confirmation of comms, including alternative methods (ie: runners, phones).		Brief description of the training activities that students will do IOT achieve the Lesson Objective. Ratio of students to instructor for practical training • What is given to trainees; • What they will do; • What the product is (i.e. how do you know when the activity is done); • What type of feedback given to trainees (self / peer / instructor) **Trg Resources:** BDCS printout, DC board, instructor has standardized solution for SB brief	

55. Test Details: EO 002.02 will be tested in EC 002.02 and PC 002.

EC 002.0*2*: Brief team members on DC considerations

EC Time: EC 20 min per trainee, includes assessor debrief.

ASSESSMENT ACTIVITY: Candidate to brief team members on various duties and considerations during damage control events. Key areas to included are, the route, hazards, equipment to be utilized, when to go on air, electrical power considerations, fitted systems, boundary and other locations and ensure status of, and confirmation of comms with teams.

Candidate may create and use an Aide-Memoire (e.g. Duty Tech style index card)

ASSESSMENT RESOURCES:
Scenarios handouts, BDCS snapshot, D/C board, Risk portfolio, assessor (to roleplay SB2IC)

IMPACT OF FAILURE: If candidate is unsuccessful, retrain and then retest.

MAR TECH RQ – PO2 EC 002.02 Briefing team members on DC considerations			
Candidate Name:	**Assessor Name:**	**Date:**	**Course session:**

INVIGILATOR NOTES:

1. Formative assessment activity
2. Use of Aide-Memoire is authorized
3. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all critical checklist items & 80% overall
4. IMPACT OF FAILURE: Remedial Training and Re-test

ITEM	UNSAT	SAT	COMMENTS
1. Brief team members on DC Considerations			
Route (Command Considerations Transit options) (*Critical)			
Hazards (*Critical)			
Equipment			
Power considerations (*Critical)			
Boundaries (*Critical)			
Fitted Systems (*Critical)			
Communications			
Locations of other teams			
Resources available			
Review ATL milestones			
Issue keys as required			

Briefing Note - ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.03 Coordinate a Section Base Team in Actions and Emergency Stations

56. Performance. Coordinate a section base team in actions and emergency stations.

57. Conditions.

a. Given:

(1) BDCS console/DC board;

(2) References;

(3) PRC with headset/telephone;

(4) Supervision;

(5) Role-player (2IC); and

(6) DC Scenario.

b. Denied: Assistance.

c. Environmental:

(1) Classroom; or

(2) BDCS Trainer.

58. Standard. The trainee shall coordinate a section base team in actions and emergency stations by:

a. Plotting on BDCS to include:

(1) Logging in using the correct credentials;

(2) Using the correct symbols; and

(3) Using the correct DC nomenclature.

b. Performing as Section Base I/C during a single DC event to include:

(1) Establishing comms;

(2) Confirming ship's status on BDCS;

(3) Reporting when Section Base is closed up;

(4) Briefing Section Base 2 I/C;

(5) Updating HQ1 on relevant DC developments; and

(6) Managing Section Base personnel.

59. Time Required.

a. EO – 1800 minutes to deliver training;

b. EC – 300 minutes for testing/quizzes; and

c. PC – 750 minutes for PC / retests.

60. Remarks. Nil.

61. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: ***At the end of this lesson, trainees will be able to plot on BDCS.***			
Basic knowledge of BDCS – 1. Log-in permissions. 2. Review of plotting symbology. 3. DC nomenclature.	A2, A3, A4, A5, A6, A7, A8	**Method / Media:** Practice plotting on BDCS software **Trg Environment:** BDCS Trainer **Activity Details:** Trainees to be provided scenario and plot relevant info on BDCS. **Trg Resources:** BDCS trainer, pre-existing scenarios.	200 min
LESSON OBJECTIVE: ***At the end of this lesson, trainees will be able to perform as section base IC during any DC event.***			
1. Establish Comms. 2. Check ships status. 3. Confirm section base team "closed up". 4. Report to HQ1. 5. Brief SBT 2IC on event and plan: a. Situation. b. Dress (ready/flood team). c. Route, hazards, etc. d. Power isolation. e. Primary/secondary zone. f. Other events. 6. Ensure boundaries established. 7. Management of comms with attack/flood teams.	A2, A3, A4, A5, A6, A7, A8	**Method / Media:** Slow time practical BDCS exercise **Trg Environment:** BDCS trainer **Activity Details:** Trainees to conduct slow time, practice scenarios to assist in developing competencies. **Trg Resources:** BDCS trainer, risk portfolio, assistance	1600 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
8. Update reports to HQ1 As DC events and personnel change: a. Event Milestones. b. Accountability of pers (to/from SBTs). c. Concurrence reports. d. Recommendations. 9. Welfare of personnel: a. Hydration. b. Cycling. c. Exhaustion. d. Heat stress. e. Hyperthermia. 10. Resource management. 11. Importance of positive control of section base.			

62. Test Details: EO 002.03 will be tested in EC 002.03 and PC 002.

EC 002.03: Coordinate a section base team in actions and emergencies

EC Time:30 min / trainee

ASSESSMENT ACTIVITY: Candidate will be Section Base I/C for a single random DC scenario (fire or flood), coordinating a section base team as the scenario develops in order to support the ship's DC organization.

Candidate may create and use an Aide-Memoire (e.g. Duty Tech style index card)

ASSESSMENT RESOURCES:

- DC boards (to match BDCS trainer);
- BDCS trainer (standalone or DCTF);
- PRCs with headsets (only if trainer unavailable);
- DC-based scenarios;
- Alternate comms (telephones, PRCs);
- Dedicated HQ1 role-player (Snr Fire or Flood Instructor; QL6A qualified legacy HTech); and
- Role-players (candidates not presently being assessed).

IMPACT OF FAILURE: If candidate is unsuccessful, retrain and then retest.

MAR TECH RQ – PO2 EC 002.04 Coordinate a SBT in actions and emergencies			
Candidate Name:	**Assessor Name:**	**Date:**	**Course session:**

INVIGILATOR NOTES:

1. Formative assessment activity.
2. Use of aide-memoire is authorized.
3. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items.
4. IMPACT OF FAILURE: Remedial Training and Re-test.

ITEM	UNSAT	SAT	COMMENTS
1. Candidate Initial Actions			
Candidate to Establish Communications			
Check of ships status: ✓ Fire pumps and pressure ✓ Ventilation ✓ Activation of fitted systems			
Acknowledge location of damage			
Confirm SBT closed up (thru 2IC) ✓ 2 DC roundsperson dispatched ✓ 4-man attack team (dressed in FF Gear)			
Report to HQ1 SBT closed up < 6 minutes			
2. Direction to Section Base			
Support command priorities (AWW/UWW/DCSS) ✓ Red openings/alpha doors/values ✓ Yankee/Zulu openings ✓ Fire/flood teams			
Confirm 2IC brief to attack/flood teams ✓ Route to incident ✓ Hazards ✓ Power considerations ✓ Restrictions			
Effective use of personal ✓ Boundaries established/report ✓ Attack/flood teams on scene ✓ Section base numbers ✓ Personnel from other locations ✓ Balancing needs with capabilities of team members			
Considers welfare of section base personal ✓ Hydration ✓ Exhaustion ✓ Team cycling (Air timings)			

ITEM	UNSAT	SAT	COMMENTS
Provide continuous reports to HQ1 status ✓ Personnel ✓ Equipment ✓ DC events			
Maintain control of BDCS/Damage control boards			
Maintain accountability of personnel thru 2IC			
Maintain proper resources to achieve command priority ✓ Section base equipment ✓ Filling of air bottles ✓ Personnel (limited #s available)			
3. Leadership			
Gather info from numerous sources			
Execute plan in support of HQ1			
Initiate participation in HQ1 requirements ✓ Offers constructive suggestion to support larger picture			
Maintain confidence and composure throughout event			
Maintain control of section base throughout event(s) ✓ Exercise positive of SBT personnel ✓ Ensure 2IC knows his place			
Provide constructive feedback to SBT upon completion of event (debrief)			

Briefing Note - ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 002: Act as Section Base I/C

PC Time: 75 mins/candidate

ASSESSMENT ACTIVITY: Candidate will be Section Base I/C for a complex, random DC scenario consisting of multiple events (action stations, emergency stations, etc.), coordinating a section base team as the scenario develops in order to support the ship's DC organization.

Communications/Reporting

Personnel not being assessed shall act as SB 2I/C in support of candidate being assessed. Will be given errors by assessor. (IE: directions for attack team to transit over fire)

Candidate may create and use an Aide-Memoire (e.g. Duty Tech style index card)

ASSESSMENT RESOURCES:

- DC boards (to match BDCS trainer);
- BDCS trainer (standalone or DCTF);
- PRCs with headsets (only if trainer unavailable);
- DC-based scenarios;
- Alternate comms (telephones, PRCs);
- Dedicated HQ1 role-player (Snr Fire or Flood Instructor; QL6A qualified legacy HTech); and
- Role-players (candidates not presently being assessed).

IMPACT OF FAILURE: If candidate is unsuccessful, retrain and then retest; TRB to be conducted IAW NPTORD 4500-54 on failure of second attempt.

MAR TECH RQ PO2 PC 002 – Act as Section Base I/C			
Candidate Name:	**Assessor Name:**	**Date:**	**Course session:**

INVIGILATOR NOTES:

1. Summative assessment activity.
2. Use of aide-memoire is authorized.
3. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items.
4. IMPACT OF FAILURE: Remedial Training and Re-test (TRB Process IAW NPTORD 4500-54).

ITEM	UNSAT	SAT	COMMENTS
1. Candidate Initial Actions			
Candidate to Establish Communications			
Check of ships status: ✓ Fire pumps and pressure ✓ Ventilation ✓ Activation of fitted systems			
Acknowledge location of damage			
Confirm SBT closed up (thru 2IC) ✓ 2 DC roundsperson dispatched ✓ 4-man attack team (dressed in FF Gear)			
Report to HQ1 SBT closed up < 6 minutes			
2. Direction to Section Base			
Support command priorities (AWW/UWW/DCSS) ✓ Red openings/alpha doors/values ✓ Yankee/Zulu openings ✓ Fire/flood teams			
Confirm 2IC brief to attack/flood teams ✓ Route to incident ✓ Hazards ✓ Power considerations ✓ Restrictions			
Effective use of personal ✓ Boundaries established/report ✓ Attack/flood teams on scene ✓ Section base numbers ✓ Personnel from other locations ✓ Balancing needs with capabilities of team members			

ITEM	UNSAT	SAT	COMMENTS
Considers welfare of section base personal ✓ Hydration ✓ Exhaustion ✓ Team cycling (Air timings)			
Provide continuous reports to HQ1 status ✓ Personnel ✓ Equipment ✓ DC events			
Maintain control of BDCS/Damage control boards			
Maintain accountability of personnel thru 2IC			
Maintain proper resources to achieve command priority ✓ Section base equipment ✓ Filling of air bottles ✓ Personnel (limited #s available)			
3. Leadership			
Gather info from numerous sources			
Execute plan in support of HQ1			
Initiate participation in HQ1 requirements ✓ Offers constructive suggestion to support larger picture			
Maintain confidence and composure throughout event			
Maintain control of section base throughout event(s) ✓ Exercise positive of SBT personnel ✓ Ensure 2IC knows his place			
Provide constructive feedback to SBT upon completion of event (debrief)			

Briefing Note - ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)	**UNSAT**	**SAT**

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 **PO 003 Adhere to Environmental, Health & Safety Regulations**

63. Performance. Adhere to environmental, health and safety regulations.

64. Conditions.

    a. Given:

        (1) References;

        (2) DWAN Access;

        (3) Supervision;

        (4) Assistance; and

        (5) Access to repair facility.

    b. Denied: Nil.

    c. Environmental:

        (1) Day or night;

        (2) At sea or alongside (trial dependent); and

        (3) HMC Ship or ashore.

65. Standard. The MAR TECH Supervisor shall adhere to environmental, health and safety regulations in support of engineering departmental requirements, IAW refs, by:

    a. Managing Captain's Ship's Book by:

        (1) Reviewing material certification documentation (AFT0480);

        (2) Organizing material certification documentation (AFT1010); and

        (3) Updating material certification documentation within the Captain's Ship's Book.

    b. Enforcing adherence to SEMS by:

        (1) Supervising implementation of environmental regulations (UT0310); and

        (2) Supervising Lockout/Tagout (LOTO) systems and equipment (UT0320).

66. References. A6 – A21.

67. Task, Skill, Knowledge Numbers. AFT0480, AFT1040, UT0310, UT0320.

68. Training Limitations:

    a. Trainees require access to DWAN workstation;

b. Training requires real LOTO equipment and documentation this could be covered by a tour and/or directed OJT;

c. Guest speaker from FTA/MatCert to be scheduled;

d. Guest speaker MS ENG post HOD to be scheduled; and

e. Guest speaker from FSE to be scheduled.

EO 003.01 Manage captain's ship's book (engineering)

69. Performance. Manage captain's ship book (engineering).

70. Conditions.

    a. Given:

        (1) References;

        (2) Workstation, with DWAN access;

        (3) Trials agendas; and

        (4) Demo of ship's OpSched.

    b. Denied: Assistance.

    c. Environmental: Classroom.

71. Standard. The trainee shall manage the Captain's Ship's Book (Engineering) by:

    a. Reviewing material certification documentation (AFT0480);

    b. Organizing material certification documentation (AFT1010); and

    c. Updating material certification documentation within the Captain's Ship's Book.

72. Time Required:

    a. EO – 100 minutes to deliver training;

    b. EO – 150 minutes guest lecturers; and

    c. EC – 200 minutes for testing/quizzes.

73. Remarks. Nil.

74. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: ***At the end of this lesson, trainees will be able to explain the contents and purpose of the Captain's Ship's Book (Engineering)***			
1. Purpose of Captain's Ship's Book; 2. Contents of Captain's Ship's Book; and 3. Explain MBS (Material Baseline Standard) - 'safe-at-sea'.	A6, A7, A8,	**Method / Media:** Interactive lecture, delivered by SME from Fleet Technical Authority (FTA) **Trg Environment:** Classroom. **Activity Details:** SME from FTA (any).	100 min guest speaker

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		Trg Resources: • Projector; and • DWAN workstation.	
LESSON OBJECTIVE: ***At the end of this lesson, trainees will be able to organize the Captain's Ship's Book (Engineering)***			
1. Validate (review and complete) contents of the Captain's Ship's Book (Engineering) (expiration of documents); 2. Trials agenda definition/contents; and 3. Location of trials agenda (F4ENG website/FTA).	A9, A10, A11.	**Method / Media:** Instructor-led walkthrough during which each candidate will have a Captain's Ship's Book (Engineering). The instructor will guide the candidates through key elements of the book. **Trg Environment:** Classroom. **Activity Details:** Have a MS ENG post HOD speak briefly to the course, outlining their expectations in terms of maintenance and upkeep of the Captain's Ship's Book. **Trg Resources:** • A copy of the Captain's Ship's Book (Engineering) for each candidate; • DWAN workstation (laptop); and • TRP schedule (sample).	100 min 50 min Guest speaker

75. Test Details: EO 003.01 will be tested in EC 003.01 and PC 003.

EC 003.01: Managing captain's ship's book (engineering).

EC Time: 200 mins

ASSESSMENT ACTIVITY: Written assessment where the Candidate will be provided with a Captain's Ship's Book (Engineering) for review (expired certificates, obsolete certificates) and a ship's readiness program. FTA has recommended to the MSEO that the department review the Captain's Ship's Book (Engineering). The PO1 has delegated this task to the candidate. The candidate will identify discrepancies that conflict immediately or during deployment and correct these errors. On completion, the candidate will brief the assessor (acting as the departmental PO1) via briefing note on their process and recommendations.

ASSESSMENT RESOURCES:

- Ship's Captain's Book of Engineering (containing expired certificates);
- DWAN access;
- Trials agendas (electronic); and
- Ship's operational schedule.

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH RQ PO2 EC 003.01: Managing Captain's Ship's Book (Engineering)			
Candidate Name:	**Assessor Name:**	**Date:**	**Course session:**

INVIGILATOR NOTES:

1. Formative assessment activity.
2. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all critical checklist items & 60% overall.
3. IMPACT OF FAILURE: Remedial Training and Re-test.

ITEM	UNSAT	SAT	COMMENTS
1. Review Captain's ships book for completeness			
Identify errors within the book (*Critical)			
Cross-Reference OPSKED against certification documentation			
Review trials agenda against OPSKED to determine feasibility of planned work			
Develop recommendations based off of findings			
Capture findings and recommendations in a briefing note (*Critical)			

Briefing Note - ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 003.02 Enforce adherence to SEMS

76. Performance. Enforce adherence to SEMS.

77. Conditions:

    a. Given:

        (1) References; and

        (2) Workstation, with DWAN access.

    b. Denied: Assistance.

    c. Environmental: Classroom.

78. Standard. The trainee shall enforce adherence to SEMS, IAW refs, by:

    a. Supervising on environmental regulations;

    b. Supervising lockout/tagout (LOTO) program.

79. Time Required.

    a. EO – 300 minutes to deliver training;

    b. EC – 200 minutes for testing / quizzes; and

    c. PC – 200 minutes for PC / retests.

80. Remarks.

    a. Ideally candidates will be able to tour a ship and audit their LOTO logs. The focus should be on identifying what their role as a PO2 managing the program as opposed to conducting a LOTO which should have been covered in previous training.

    b. This training assumes that the LOTO coordinator on ship will normally be filled by an Electrician who has received additional LOTO training on the USQ PO2 Electrical.

81. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: At the end of this lesson, trainees will be able to supervise on implementation of environmental regulations.			
1. Authority for SEMS; 2. Contents of SEMS;	A12, A13, A14, A15,	**Method / Media:** Guest lecture SME on environmental health and safety (SEMS).	100 min Guest Speaker

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
3. Legal implications of SEMS adherence; 4. Definition of 'due diligence'; 5. Commanding Officer's expectations; 6. Safety and environmental programs on ship; 7. Safety and environmental requirements (rounds, etc.); 8. Following up on events (CF-98; DND 663); and 9. Environmental considerations (fuel, black water, oil transfer, POL, HAZMAT, MSDS, WHMIS, Ozone Depleting Substances (ODS), Controlled Substances (i.e. CTAT substance in towed array)).	A16, A17, A18, A19, A20, A21,	**Trg Environment:** Classroom. **Activity Details:** SME lecture (MSEO, safety officer (Formation Safety and Engineering), FMF personnel, Base/Formation Environmental Officer(ideal)): • Examples of actual charges laid (by Govt of Canada); • Examples of active and healthy environmental/safety program (consistent, proper monitoring, buy-in, what 'right' looks like); and • Case study (examination of an event that went well/badly, ie. HMCS Calgary fuel spill weekend of 24 Feb 18). **Trg Resources:** e.g. Handouts (CF-98, DND 663) DWAN access (SEMS) Projector (for SME presentation)	50 Min
LESSON OBJECTIVE: ***At the end of this lesson, trainees will be able to supervise lockout/tagout program.***			
1. Review LOTO best practices a. Types of lockouts (i.e. 440 lockouts); b. Effective use of lockout/tagout gear (i.e. no requirement for chains on all systems); c. Liability and responsibility; and d. What is the role of the PO2 in the LOTO program on ship. 2. Lockout/tagout program: a. Legislation (NAVORD, Canada Labour Code, civilian management); b. Deficiencies;	A13, A14, A15, A16.	**Method / Media:** Interactive lecture, Tour. **Trg Environment:** Classroom. **Activity Details:** Interactive lecture containing photos of assorted LOTO equipment. Candidates to receive samples of completed documentation (clearly indicated, possibly redacted ship information). **Trg Resources:** Pictures of: • Sample lockbox with locking mechanisms (multi-lock); • Clamshells;	50 min 100 min Tour

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Paperwork (shipboard lockouts, FMF/RF lockouts, hull transfers, tag control); d. Authority to remove locks and tags; and e. Document management. 3. Tour to Audit a LOTO program a. Roles and responsibilities b. Common infractions		• Chains; • Bars; • Assorted lockout/tagout tags; • DWAN access; and • Projector.	

82. Test Details. EO 003.02 will be tested in EC 003.02 and PC 003.

EC 003.02: Enforce adherence to SEMS

EC Time: 200 min

ASSESSMENT ACTIVITY: Candidate will answer a series of questions on two scenarios, one on a SEMS infraction, and the other on a LOTO infraction. Possible scenarios include forms missing signatures, forms not dated, locks left behind, paperwork not completed, duplicate paperwork, lockout/tagout resource concerns; fuel spills, effluent discharge, oil transfers, ammunition transfer.

The exam will be open book. The candidate will be required to identify specific references.

Scenario 1: SEMS

1. Operational/Exercise status of the unit (where is the ship?).
2. Ships Readiness I.E. Ready Duty Ship.
3. Example Safety or Environmental Scenario.

Questions could include:

1. What was the infraction?
2. What is the safety or environmental impact?
3. What is the follow-on safety actions? Example no smoking throughout ship…
4. Who do you report this incident to? (Water vs land)
5. Who would respond to the applicable event?
6. What actions would you take to correct each infraction?
7. What is the clean-up / containment plan?

Scenario 2: LOTO

Scenario could include:

4. Operational/Exercise status of the unit (where is the ship?).
5. Ships Readiness I.E. Ready Duty Ship.
6. Example LOTO book.
7. Completed LOTO form.
   a. An improper description of the item being locked out (i.e. valve vs specific valve).
   b. An improper authority for approval.
   c. Improper authority for reinstatement.

Questions could include:

1. What lock would you use?
2. What LOTO infraction(s) do you see?
3. Who has the authority to cut the lock?
4. Would you get the lock cut off?
5. How would you get the lock cut off?
6. What actions would you take to correct each infraction?

ASSESSMENT RESOURCES:

- DWAN Workstation (access to electronic resources)
- Test booklets

PC 003: Adhere to environmental health & safety regulations

PC Time: 200 mins; all trainees assessed concurrently, and on same scenario

ASSESSMENT ACTIVITY: In-basket activity (scenario with a number of scheduled inputs/events that the candidate will have to manage over the conduct of the scenario). The overall scenario will be related to Environmental Health and Safety Regulations adherence, with the candidate being assessed on managing a SEMS infraction, lock-out/tag-out infraction or issue, or an issue related to certification documentation. Scheduled inputs (number of which will be based upon the scenario; 6-10; ranging in difficulty from simple to complex) are to be relevant EH&S issues that could be expected to arise during the events that unfold. 3-4 key inputs should directly relate to SEM infractions, remainder do not need to be.

The candidate will be assessed on both their actions taken, as well as their ability to prioritize and address these concerns. References will be provided, but the individual time allocated will be insufficient for a candidate to read the reference in its entirety to be successful in completion of the scenario.

On completion, the candidate will be required to capture the events that transpired during the scenario (format unimportant, but some kind of 'briefing note' will need to be produced).

ASSESSMENT RESOURCES:

- Trainee DWAN account (initial course admin);
- Electronic references (SEMS, NEM, NaMMS, current SSOs);
- Engineering Book of the Captain's Ship's Books (hard copy);
- DWAN workstation for each candidate;
- Projector;
- PowerPoint;
- Scenario (grounded in reality; derived from Periodic Engineering Letters/Lessons Learned);
- Standardized solutions for each scenario (based on SME input); and
- Classroom.

IMPACT OF FAILURE: TRB Process IAW NPTORD 4500-54

MAR TECH RQ PO2 PC 003 – Adhere to Environment Health and Safety Regulations			
Candidate Name:	**Assessor Name:**	**Date:**	**Course session:**

INVIGILATOR NOTES:

1. Summative assessment activity.
2. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all critical checklist items & 60% overall.
3. IMPACT OF FAILURE: Remedial Training and Re-test (TRB Process IAW NPTORD 4500-54).

ITEM	UNSAT	SAT	COMMENTS
1. Prepare an AAR on an Environmental Health and Safety Infraction			
Identify the infraction(s) in the initial scenario (*Critical)			
Identify the correct reference(s) that detail the infraction (to the para) (*Critical)			
Determining the potential cause(s) of the infraction(s)			
Actions taken to minimize environmental/health/safety impact			
Follow-on actions (COAs) to prevent or mitigate repeat of event in future			

Briefing Note - ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 004 **PO 004 Coordinate Marine Systems Maintenance**

83. Performance. Coordinate marine systems maintenance.

84. Conditions:

    a. Given:

        (1) References;

        (2) DWAN Access;

        (3) DRMIS;

        (4) Supervision;

        (5) Assistance; and

        (6) Access to repair facility.

    b. Denied: Nil.

    c. Environmental:

        (1) Day or night;

        (2) At sea or alongside (trial dependent); and

        (3) HMC Ship or ashore.

85. Standard. The MAR TECH Supervisor shall coordinate marine systems maintenance in support of ship's program, IAW refs, by:

    a. Coordinating departmental maintenance alongside and at sea (maintenance watch IC), to include:

        (1) Organizing work parties (availability, people, equipment, parts, time);

        (2) Prioritizing Corrective and Planned Maintenance, including work periods;

        (3) Reviewing preventative maintenance schedule (AFT0520);

        (4) Reviewing particularized maintenance and repair schedule (PMRS) (AFT0490);

        (5) Reviewing ship's standardized maintenance and repair schedule (SSMRS) (AFT0530);

        (6) Liaising with other sections and departments when required;

        (7) Completing work acceptance form (AFT1040);

        (8) Presenting maintenance portfolio; and

        (9) Drafting technical documentation (AFT1310) to include:

i. Preparing pre-refit survey reports (650s) (AFT0330);

ii. Drafting engineering change requests (AFT1240);

iii. Completing trial documentation; and

iv. Drafting operational deficiency reports (AFT1270).

b. Coordinating marine engineering trials (AFT1130) to include:

(1) Reviewing work description inspection test plans (AFT0580);

(2) Reviewing test and trial proposals (AFT0550);

(3) Completing trial acceptance forms (AFT1030);

(4) Analyzing inspections/tests/trials, results (AFT0880);

(5) Reviewing test and trial reports (AFT0560); and

(6) Actioning quality assurance defect reports (AFT0700).

c. Brief HOD and CHOD.

86. References. A6, A7, A8, A12, A22, A23, A24, C1, C2, C3

87. Task, Skill, Knowledge Numbers. AFT0520, AFT0490, AFT0530, AFT1040, AFT1310, AFT0330, AFT1240, AFT1270, AFT1130, AFT0580, AFT0550, AFT1030, AFT0880, AFT0560, AFT0700.

88. Training Limitations.

a. Requires DWAN workstation for trainees.

EO 004.01 Coordinate Departmental Maintenance

89. Performance. Coordinate departmental maintenance.

90. Conditions.

b. Given:

(1) References;

(2) Workstation, with DWAN access;

(3) Work description inspection test plans;

(4) Test and trial proposals;

(5) Test acceptance form;

(6) Inspection/test/trial results;

(7) Test/trial reports;

(8) Quality assurance report; and

(9) Scenario.

c. Denied: Assistance.

d. Environmental: Classroom.

91. Standard. The trainee shall coordinate departmental maintenance, IAW refs by:

a. Coordinating departmental maintenance alongside and at sea (maintenance watch IC), to include:

(1) Organizing work parties (availability, people, equipment, parts, time);

(2) Prioritizing Corrective and Planned Maintenance, including work periods;

(3) Reviewing preventative maintenance schedule (AFT0520);

(4) Reviewing particularized maintenance and repair schedule (PMRS) (AFT0490);

(5) Reviewing ship's standardized maintenance and repair schedule (SSMRS) (AFT0530);

(6) Liaising with other sections and departments when required;

(7) Completing work acceptance form (AFT1040);

(8) Presenting maintenance portfolio; and

(9) Drafting technical documentation (AFT1310) to include:

i. Preparing pre-refit survey reports (650s) (AFT0330);

ii. Drafting engineering change requests (AFT1240); and

iii. Drafting operational deficiency reports (AFT1270).

92. Time Required.

a. EO – 700 minutes to deliver training; and

b. EC – 100 minutes for testing/quizzes.

93. Remarks. Nil.

94. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: ***At the end of this lesson, trainees will be able to coordinate departmental maintenance***			
1. Planned Maintenance Review Schedule.; 2. Types of work periods; 3. Determining the scope of work; 4. Ship's Standardized Maintenance and Repair Schedule (SSMRS); 5. Defining NWs (type of work order; outside of ship's scope PM); 6. Contents of the Ten Week Message, including priorities; 7. Inputs into the Eight Week Message (what section/dept contributes): a. NEMTIs; and b. Group prioritization and justification. 8. Drafting technical documentation (AFT1310) to include: a. Preparing pre-refit survey reports (650s) (AFT0330); b. Drafting engineering change requests (AFT1240); and	A2, A12, A22, A23, C1, C2, C3	**Method / Media:** Syndicate work; 2 groups (Mech and Elec, each with an instructor and a list of 100 'dept priorities' – 50 mech, 50 elec). Each group will work collaboratively to determine a list of priorities for their section, and then both syndicates (sections) will participate in a debate activity that represents the collaborative prioritization of dept/ship maintenance priorities. **Trg Environment:** Classroom. **Activity Details:** There will be two guest speakers prior to the conduct of the syndicate activity: 1. Ship Support Officer (SSO) will provide context on what happens in terms of maintenance and maintenance priorities external to the ship including 650s; and 2. FTA & FEO-2 Rep to discuss ECs and Opdefs including but not limited to: a. Purpose; b. References; c. Processing; and d. Best practices. **Trg Resources:**	100 min guest lecturers 400 min syndicate work 200min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Drafting operational deficiency reports (AFT1270).		• Lists of section/dept priorities for syndicate work; and • Prebuilt templates (outstanding DRMIS work orders).	

95. Test Details: EO 004.01 will be tested in EC 004.01 and PC 004.

EC 004.01 Coordinate marine systems maintenance

EC Time: 100 min

ASSESSMENT ACTIVITY: Scenario-based; candidate receives a 'Ten Week Message,' outlining all the outstanding work the ship requires prior to going into a work period. The candidate will vet and prioritize all respective mechanical/electrical requirements.

ASSESSMENT RESOURCES:

- Trainee DWAN account (initial course admin)
- Electronic references (Main Propulsion Manual, NEM, NaMMS, current SSOs)
- Engineering Book of the Captain's Ship's Books (hard copy)
- NEMTI
- DWAN workstation for each candidate
- Projector
- PowerPoint
- Scenario (grounded in reality; derived from ship's Tiered Readiness Program (TRP) and PMRS, initiated by Ship's Supporting Officer sending 'Ten Week Message')
- Classroom
- Work description inspection test plans
- Test and trial proposals
- Test acceptance form
- Inspection/test/trial results
- Test/trial reports
- Quality assurance report (negative result)

IMPACT OF FAILURE: Remedial Training.

MAR TECH RQ PO2 EC 004.01 Coordinate departmental maintenance			
Candidate Name:	**Assessor Name:**	**Date:**	**Course session:**

INVIGILATOR NOTES:

1. Formative assessment activity.
2. PASS / FAIL CRITERIA: Trainees must achieve a 60% overall.
3. IMPACT OF FAILURE: Remedial Training and Re-test.

ITEM	UNSAT	SAT	COMMENTS
1. Interpret 10 Week work package message			
Identify sectional work notifications (Mech/Elec)			
Prioritize notification in order of importance			
2. Prepare for 8 Week work package message			
Participate in intra-departmental priority meeting			
Create a master prioritized notification list			
3. Work as a team member			
Preparedness for meeting			
Respectfulness			
Follows functioning guidelines (FLOOR model)			

Briefing Note - ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 004.02 Coordinate marine engineering trials

96. Performance. Coordinate marine engineering trials.

97. Conditions.

    a. Given:

        (1) References;
        (2) Workstation, with DWAN access;
        (3) Work description inspection test plans;
        (4) Test and trial proposals;
        (5) Test acceptance form;
        (6) Inspection/test/trial results;
        (7) Test/trial reports;
        (8) Quality assurance report; and
        (9) Scenario.

    b. Denied: Assistance.

    c. Environmental: Classroom.

98. Standard. The trainee shall coordinate marine engineering trials, IAW refs, by:

    a. Reviewing work description inspection test plans (AFT0580);

    b. Reviewing test and trial proposals (AFT0550);

    c. Organizing marine systems trials;

    d. Completing trial acceptance forms (AFT1030);

    e. Analyzing inspections/tests/trials results (AFT0880);

    f. Actioning quality assurance defect reports (AFT0700); and

    g. Completing post-trials requirements.

99. Time Required:

    a. EO – 300 minutes to deliver training;

    b. EC – 100 minutes for testing/quizzes; and

    c. PC – 400 minutes for PC / retests.

100. Remarks. Nil.

101. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: At the end of this lesson, trainees will be able to organize marine systems trials.			
1. Types of trials (i.e. Certification Trials); 2. Requirements for trials; 3. Personnel involved; 4. Trial authority (conducting and convening authorities; vary from trial to trial); and 5. Logistics considerations (location, time of year, ship's evolutions, personnel available).	A11, A12, A22	**Method / Media:** Interactive lecture. **Trg Environment:** Classroom. **Activity Details:** **Guest lectures** • FMF Trial Officer/Chief; • FTA SME; and • FER-2. **Trg Resources:** • IT (projector, DWAN workstation, PowerPoints).	200 min
LESSON OBJECTIVE: At the end of this lesson, trainees will be able to complete post-trials requirements			
1. Review a completed trials report; 2. Trial scheduling process (corresponding with key FTA personnel); 3. Waivers process (SME; provision of input to PO1/EO for the purpose of briefing the CO); and 4. Risk assessment process.	A6, A7, A8, A24	**Method / Media:** Lecture on relevant teaching points**,** followed by two case studies (one mech, one elec; on a 'historical'/factual failed trials report that requires a risk assessment/waiver, with time restrictions for retrial). **Trg Environment:** Classroom. **Activity Details:** • Each candidate will have a copy of the report (and relevant templates, ie. FTA risk assessment template) but not the results • The course will divide into groups of no more than six • Instructor to guide each group of candidates through follow-on actions and administrative requirements using the example provided **Trg Resources:** • IT (projector, DWAN workstation, PowerPoints);	100 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• Case studies (one mech, one elec); and • Report templates.	

102. Test Details: EO 004.02 will be tested in EC 004.02 and PC 004.

EC 004.02: Coordinate marine engineering trials

EC Time: 100 mins

ASSESSMENT ACTIVITY: End of work period, candidate has to determine what trials need to be completed prior to proceeding to sea. Each candidate will write an exam during which they will answer questions based on a scenario. The candidate will have to determine what trials will need to be conducted, who would conduct the trials, follow-on activities (completing the trial acceptance forms, analyzing results, reviewing test/trial reports, risk assessment, waivers, and formulating a plan to action follow-on repairs and maintenance).

Scenario 1:Lifting appliance certification fails prior to planned deployment

1. Operational/Exercise status of the unit (where is the ship?)
2. Ships Readiness I.E. Ready Duty Ship
3. Length of testing for certification (test/trial agenda)

Questions could include:

1. What options exist to achieve certification (replace/TranReq)?
2. What was the mission impact?
3. Who is the technical authority for the testing?
4. How long is certification valid?
5. What is the procedure for recertification?
6. What restrictions are governed within the recertification process?
7. Is deploying on a waiver an option?
8. Who is required to accept the risk?
9. Who are the stakeholders that should be notified of failure?

ASSESSMENT RESOURCES:

- Sample reports;
- Trial agenda; and
- Report templates (trial acceptance form).

IMPACT OF FAILURE: Remedial Training.

PC 004: Coordinate marine systems maintenance

PC Time: 200 mins (independent vetting and prioritization and group prioritization); 100 mins (for review); 100 minutes (QA report)
PC Time required in mins (account for each trainee, include student debrief time, account for retests)

ASSESSMENT ACTIVITY: Scenario-based; candidate receives a 'Ten Week Message,' outlining all the outstanding work the ship requires prior to going into a work period. The candidate will vet and prioritize all respective mechanical/electrical requirements, attend the 'Eight Week Message' Meeting, collaborate with other candidates to determine the 'ship's' repair priorities, and then draft an 'Eight Week Message' IAW refs.

On completion, each candidate will receive a trial agenda (for a common trial that would follow a work period). When trials are 'complete,' the candidate will review work description inspection test plans IAW NEMTI, review test and trial proposals, complete the test acceptance form, analyze inspection/test/trial results, review test and trial reports, and action quality assurance report (input from external authority; test result comes back negative for a 'safe at sea' requirement). (Mech and elec specific; DG-Set may be a good choice).

Candidate will be required to complete

Items will be prioritized High, Medium, or Low for assessment purposes.

Assessor will play the part of Engineering Officer/CEng/Dept Coordinator

ASSESSMENT RESOURCES:

- Trainee DWAN account (initial course admin);
- Electronic references (Main Propulsion Manual, NEM, NaMMS, current SSOs);
- Engineering Book of the Captain's Ship's Books (hard copy);
- NEMTI;
- DWAN workstation for each candidate;
- Projector;
- PowerPoint;
- Scenario (grounded in reality; derived from ship's Tiered Readiness Program (TRP) and PMRS, initiated by Ship's Supporting Officer sending 'Ten Week Message');
- Classroom;
- Work description inspection test plans;
- Test and trial proposals;
- Test acceptance form;
- Inspection/test/trial results;
- Test/trial reports; and
- Quality assurance report (negative result).

IMPACT OF FAILURE: TRB Process to be conducted IAW NPTORD 4500-54.

MAR TECH RQ PO2 PC 004 – Coordinate Marine Systems Maintenance			
Candidate Name:	**Assessor Name:**	**Date:**	**Course session:**

INVIGILATOR NOTES:

1. Summative assessment activity.
2. PASS / FAIL CRITERIA: Trainees must achieve a 60% overall.
3. IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt.

ITEM	UNSAT	SAT	COMMENTS
1. Candidate prioritized tasks effectively			
Prioritize section requirements (H/M/L) independently			
Defend section priorities at 'Eight Week Message' Meeting			
Candidate will provide input on departmental priorities at 'Eight Week Message' Meeting			
Candidate will prepare an 'Eight Week Message' based on 'Eight Week Message Meeting' IAW refs			
2. Review test/trial documentation			
Candidate will locate the error(s) in trial agenda documentation			
Write a QA report based upon the negative trial results, including recommendations to rectify the equipment/system deficiency			
3. Prepare an AAR on an Environmental Health and Safety Infraction			
The MAR TECH Supervisor will draft an AAR on an EH&S infraction to include: ✓ Identify the infraction(s) in the initial scenario ✓ Identify the correct reference(s) that detail the infraction (to the para) ✓ Determining the potential cause(s) of the infraction(s) ✓ Actions taken to minimize environmental/health/safety impact ✓ Follow-on actions (COAs) to prevent or mitigate repeat of event in future			
4. Work as a team member			
Prepared for meeting			
Respectful			
Follows functioning guidelines (FLOOR model)			

Briefing Note - ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 **PO 005 Coordinate Training and Readiness**

103. Performance. Coordinate training and readiness (AET1010).

104. Conditions.

    a. Given:

        (1) References;

        (2) Assistance;

        (3) Monitor Mass;

        (4) Watch & Station Bill;

        (5) Automatic Establishment Report; and

        (6) Workstation with DWAN access.

    b. Denied: Nil.

    c. Environmental: Any conditions, day, or night, at sea or alongside.

105. Standard. The Marine Technician Supervisor shall independently coordinate training and readiness IAW refs by:

    a. Coordinating personnel employment by:

        (1) Analyzing crewing requirements to include:

            i. Ensuring pers are scheduled for refresher training;

            ii. Tracking APRV requirements; and

            iii. Ensuring all training requirements are up to date.

        (2) Utilizing Transition Centre resources; and

        (3) Managing the Automatic Establishment Report.

    b. Managing human resources in Monitor Mass by:

        (1) Identifying leave discrepancies;

        (2) Employing training tracker;

        (3) Analyzing parade state; and

        (4) Running queries via analytics function.

106. References. A25, A26, A27, A28, A29, A30, A31.

107. Task, Skill, Knowledge Numbers. AET1010.

108. Training Limitations.

    a. Lack of a Monitor Mass sandbox will limit trainees' ability to interact with fictional data in the program by entering data and completing queries. Training will default to demonstrations completed by instructors.

EO 005.01 Manage human resources

109. Performance. Manage human resources.

110. Conditions.

a. Given:

(1) References;

(2) Assistance;

(3) Monitor MASS trainee workbook;

(4) Monitor MASS with sandbox; and

(5) Workstation with DWAN access.

b. Denied: Nil.

c. Environmental: Classroom.

111. Standard. The trainee shall independently manage human resources in Monitor Mass IAW refs by:

a. Identifying leave discrepancies;

b. Employing training tracker;

c. Analyzing parade state; and

d. Running queries via analytics function.

112. Time Required.

a. EO – 400 mins to deliver training.

113. Remarks. Nil.

114. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Utilizing monitor mass to include but not limited to:			
1. Identifying leave discrepancies by; a. Conducting leave audits. 2. Employing training tracker; 3. Analyzing parade state; and 4. Running queries via analytics function to include: a. Qualifications; b. Operational tempo; c. Refresher training; d. Fitness testing; e. Passport; and f. Language.	A31	**Method / Media:** EDI. **Trg Environment:** Classroom. **Activity Details:** Instructor to explain the purpose of various portions of Monitor MASS, demonstrate where and how to access essential functions, and have trainees follow on their own systems. Reference to user guide when appropriate as instructor guides trainees through various scenarios. **Trg Resources:**	400 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• Monitor MASS user guide; and • Job Aide for Monitor MASS.	

115. Test Details. There will be no EC for EO 005.01. Trainees will be given ongoing feedback as they progress through this EO.

EO 005.02 Coordinate personnel employment.

116. Performance. Coordinate personnel employment.

117. Conditions.

a. Given:

(1) References;

(2) Assistance;

(3) Monitor Mass;

(4) Watch & Station Bill template;

(5) Impact statement template;

(6) Automatic Establishment Report; and

(7) Workstation with DWAN access.

b. Denied: Nil.

c. Environmental: Classroom.

118. Standard. The trainee shall independently coordinate personnel employment IAW refs by:

a. Analyzing crewing requirements ton include:

(1) Ensuring personnel are scheduled for refresher training;

(2) Tracking APRV requirements;

(3) Ensuring training requirements are up to date;

(4) Utilizing PCC services by:

i. Identifying posting types;

ii. Determining manning priorities to include the following:

a) Tracking personnel; and

b) Tracking training.

b. Utilizing Transition Centre resources; and

c. Managing the Automatic Establishment Report to include developing the Watch and Station Bill.

119. Time Required.

a. EO – 450 mins to deliver training;

b. EC – 50 mins (total for all trainees); and

c. PC – 200 mins (total for all trainees).

120. Remarks. Nil.

121. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Analyzing crewing requirements by:			
1. Utilizing PCC services by: a. Describing PCC organization structure; b. Identifying posting types to include: 1. CFTPO requirements; 2. TD; 3. AP; and 4. IR; c. Determining manning priorities to include: 1. Tracking personnel; 2. Tracking training; and 3. Impact statements. d. Drafting Personnel Shortage Message.	A27, A28, A30	**Method / Media:** GL, IL. **Trg Environment:** Classroom. **Activity Details:** PCC guest lecturer will present the information contained in the learning objectives with the class. Trainees to then be provided a scenario in which a manning shortage is present. Trainees will be required to assess the impact based on ship's mission and draft an impact statement. **Trg Resources:** • Guest lecturer; • Scenario; and • Impact statement template.	100 min Guest Lecture 100 Activity
LESSON OBJECTIVE: Utilizing JPSU resources by:			
1. Describing JPSU organization structure to include purpose of JPSU; and 2. Posting process to JPSU to include: a. Criteria; b. Employment to include return to work; and c. Support.	A26	**Method / Media:** GL/GD. **Trg Environment:** Classroom**.** **Activity Details:** Guest lecturer from JPSU to ensure that all lesson objectives are discussed in class. Prior to lecture trainees shall familiarize themselves with ref to ensure active participation in discussion. **Trg Resources:** • Guest lecturer; • Reference; and • Guided discussion setup.	100 min Guest Lecture 100 Activity
LESSON OBJECTIVE: Managing the REMAR by:			
1. Developing the Watch and Station Bill to include: a. Watch rotations; and b. Special parties' requirements.	A2	**Method / Media:** IL. **Trg Environment:** Classroom. **Activity Details:** Instructor to ensure that all relevant teaching points are covered during the lecture. **Trg Resources:** • Watch and Station bill template from ref.	50 min

7. Test Details. EO 006.02 will be tested in EC 006.02 and PC 006 IAW table below and Annex K.

EC 005.02: Coordinate personnel employment

EC Time: 40 mins (concurrent) + 10 min debrief.

ASSESSMENT ACTIVITY: Trainees will be provided a randomly selected scenario in which they will review a watch and station bill for accuracy. The watch and station bill will contain built in errors and deficiencies that the trainee will be required to identify and rectify. Trainee will draft an email to the Combat Chief (instructor) with the required information for a PSM. The scenario will consist of several of the following likely issues:

- Members on MELs
- Members in reception
- Member on leave
- Member is not at unit
- Personal information is incorrect
- Member is missing qualifications
- Member is reservist and at end of contract

Pass/fail criteria: Passing 70% of EC checklist.

ASSESSMENT RESOURCES:

- Scenario;
- DWAN workstation;
- Monitor MASS Sandbox
- Ship personnel loaded into Sandbox; and
- Monitor MASS job aide.

IMPACT OF FAILURE: Remedial Training.

RQ PO2 MAR TECH Supervisor – EC 006.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a 70% SATs on the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
3. Coordinate Personnel Employment			
Correctly identifies manning deficiencies			
Identifies qualifications required and assigns each a priority			
Conducts query in Monitor MASS for missing qualifications/personnel			
Correctly identifies personnel required			
Trainee drafts the Watch and Station Bill			
Trainee completes email required for PSM			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 005: Coordinate training and readiness

PC Time: 200 min

ASSESSMENT ACTIVITY: Scenario will be provided to trainees in which they must develop a Watch and Station Bill. Trainee will be required to utilize the Monitor MASS sandbox to follow the process of identifying manning deficiencies identifying personal to correct those deficiencies. Additionally, trainees are required to provide rational behind personnel selection choices.
Pass/fail criteria: Passing 80% taken from PC checklist.

ASSESSMENT RESOURCES:

- Scenario;
- Monitor MASS Sandbox; and
- Monitor MASS job aide.

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

RQ PO2 MAR TECH Supervisor – PC 006 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve an 80% SAT on the remainder of the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test / TRB after second unsuccessful test.

ITEM	SAT	UNSAT	COMMENTS
1. Coordinate Personnel Requirements			
Correctly identifies manning deficiencies			
Identifies qualifications required and assigns each a priority			
Conducts query in Monitor MASS for missing qualifications			
Correctly identifies personnel required			
2. Watch and Station Bill			
Trainee completes the Watch and Station Bill			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**UNSAT**	**SAT**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 006 **PO 006 Act as DRMIS Manager**

This PO will be captured by the DRMIS Maintenance Manager - 119098. MITE Code 119098 will be awarded on successful completion of RQ PO2.

All timings, teaching points and assessments will conducted IAW with the DRMIS Maintenance Manager QSP.

This PO could be completed first as the knowledge and skills covered could be leveraged to complete activities found in other POs.

ANNEX A – **MAIN REFERENCES**

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	CFCD 132 RCN Damage Control Manual
A2	RCN SSO AL11
A3	Sea Training Group Readiness Training Guide RTG 1.0
A4	C-23-045-000/AG-003 Damage Control (SEA) Vol 3 0f 3 Ship Stability
A5	C-23-HFX-000/MS-003 Manual of Trim and Stability Applicable to Halifax Class
A6	NAVORD 3001-0 In-Service Naval Materiel Risk Management - Policy
A7	NAVORD 3001-1 In-Service Naval Materiel Risk Management - Process
A8	C-23-005-000/AG-002 Naval Material Risk Management
A9	C-28-020-001/TB-001 Technical Bulletin (Marine) In-Service Certification Requirements for Shipboard Lifting Equipment Applicable to all Class of Ships
A10	C-03-055-033/AA-001 Captain's Ship Books Former HMCS Huron DDH 281
A11	C-27-778-000/NT-001 trial Agenda for Steering Agreement Applicable to Halifax Class
A12	C-03-005-033/AA-00 Naval Engineering Manual
A13	Halifax Class Safety and Environmental Management System (SEMS) Manual
A14	Kingston Class Safety and Environmental Management System (SEMS) Manual
A15	Victoria Class Safety and Environmental Management System (SEMS) Manual
A16	Harry Dewolf Class Safety and Environmental Management System (SEMS) Manual
A17	Orca Class Safety and Environmental Management System (SEMS) Manual
A18	Fisheries Act (R.S.S., 1985, c. F-14)
A19	Canadian Environmental Protection Act, 1999 (S.C. 1999, c. 33)

A20	Transportation of Dangerous goods Act, 1992 (S.C. 1992, c. 34)
A21	Canadian Environment Assessment Act, 2012 (S.C. 2012, c. 19, s. 52)
A22	C-03-005-012/AM-001 NaMMS
A23	NAVORD 3000-0 Material Baseline Standard
A24	E-003 HMCS Montreal Rudder Inspection PRMS
A25	MARPACORD 5320-2
A26	MARPACORD 5320-3
A27	NAVORD 5000-0
A28	NAVORD 4500-0
A29	CFCD 129
A30	RCN Crewing Priorities 2017-19 RDIMS#408507
A31	MCS-PERS (Monitor MASS) User guide 4.0.5 24 APR 18
C1	Uptime – Strategies for Excellence in Maintenance Management 3rd Ed
C2	Maintenance Planning and Scheduling Handbook
C3	Reliability Centered Maintenance – Reengineered; Practical Optimization of the RCM Process with RCM-R

ANNEX B – **TASK LIST**

Task No	Task Statement	T/NT	PO
AET1010	Coordinate training and readiness	T	006
AFT0330	Prepare pre-refit survey reports (650s)	T	004
AFT0340	Prepare unsatisfactory condition reports (UCR)	NT	
AFT0450	Review engineering change proposal	NT	
AFT0480	Review material certification documentation	T	003
AFT0490	Review particularized maintenance & repair Schedule (PMRS)	T	004
AFT0520	Review preventive maintenance schedule	T	004
AFT0530	Review Ship's Standardized Maintenance & Repair Schedule (SSMRS)	T	004
AFT0550	Review test and trial proposals	T	004
AFT0560	Review test and trial reports	T	004
AFT0570	Review unsatisfactory condition reports (UCR)	NT	
AFT0580	Review work description Inspection Test Plans	T	004
AFT0590	Review work request priority spreadsheet (WRPS)	T	004
AFT0700	Action quality assurance defect report	T	004
AFT0720	Action unsatisfactory condition reports (UCR)	NT	
AFT0880	Analyze inspections/tests/trials results	T	004
AFT1010	Complete material certification documentation	T	003
AFT1020	Complete Test acceptance form	NT	
AFT1030	Complete Trial acceptance form	T	004
AFT1040	Complete work acceptance form	T	004
AFT1130	Coordinate marine engineering trials	T	004
AFT1240	Drafting engineering change requests	T	004
AFT1270	Draft operational deficiency report (OpDeF)	T	004
AFT1310	Drafting technical documentation	T	004
BT0070	Inspect controllable reversible pitch propeller system	NT	
BT0080	Inspect main propulsion gearing	NT	
BT0090	Inspect main propulsion shafting	NT	
CT0320	Report on marine systems engineering parameters	NT	
CT0340	Report on propulsion deficiencies	NT	

IT0140	Coordinate on scene-damage control actions	NT	
IT0150	Coordinate section base team in action/emergencies	T	002
IT0490	Manage section base team in action/emergencies	NT	
UT0310	Supervise Implementation of environmental regulations	T	003
UT0320	Supervise lock-out/tag-out systems and equipment	T	003
	Coordinate section base team training	T	002
	Brief team members on DC considerations	T	002
	Manage Captain's Ship's Book (Engineering)	T	003
	Enforce adherence to SEMS	T	003
	Act as Section Base IC	T	002
	Supervise Implementation of Safety Regulations	T	003
	Coordinate departmental maintenance	T	004

ANNEX C – **QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS**

QSP Finalization and Review Notes and Recommendations:

For next review/board:
Areas such as stability and DC roles/responsibilities should be coordinated across RQ PO2, USQs, and RQ PO1 so they are tied together, build on previous tasks, and avoid and gaps in training (and tasks). For example, the gap from a PO2 to PO1 level with regards to stability and acting in DC positions is too large.

PO 002 requires full review and be addressed with Sea Trg and new way this is managed on Ship - NTDCP, DC School, Fleet, Sea Trg – has the job changed, if yes then this PO needs to change to reflect that.
All content should be formatively assessed. If not, it reduces student buy-in and doesn't adequately prepare a PO2 to do that aspect of the job.

Dates:

19 Feb – 02 Mar 2018

Location:

Haida Bldg rm 105C, Work Point, CFB Esquimalt

Board Members:

Rank and name	MOSID / Unit	Dates
CPO2 Y. Gagné, Mar Tech TM	00379 / NPTG HQ	Only present for small periods of time throughout board. SitReps sent every second day.
CPO2 J.P. Ayotte, Chair	00379 / NTDC(P)	19-23 Feb (absent for appointments throughout), 27 Feb
CPO2 S. Wilcox, SME	00379 /ADM Mat/DGPM	19 Feb -
CPO2 V. Prosper, SME	00379 / NFS(A)	19 Feb – 1 Mar
PO1 S. Blackbourn, SME	00379 / NTDC(P)	19-22 Feb, 26 Feb-
SLt Adrian Smart, TDO OJT	00211 / NTDC(P)	19 Feb -
SLt Sarah Valentine, TDO	00211 / NTDC(P)	19 Feb -

Board Objective:

The board's objective was to come up with realistic training objectives that match the job requirements of a PO2 MAR TECH Supervisor within the mandated time frame.

Discussions / Decisions:

**Decisions which are stricken through were changed at a later time*

Date	Decision/Event	Detail
19 Feb	Board convened	
	Intro brief	Given by SLt Smart to all board mbrs
	CFITES Brief	Given by SLt Smart to all board mbrs
	NTDC(P) / Cdr's Intent brief	Given by LCdr Zubrinich (NTDC(P) DCO/acting CO) and LCdr Day (NTDC(P) MSE Div Cdr) to all board members: • Short timeline • Switch from job to how trained • Some legacy elements but think brand new, what the occupation envisions = future • Always look at impact; strive for 70% solution (perfection will come after conduct) • TEL component; keep in mind bi-coastal; DL, simulation, 3D modeling, etc.; consider CAF organization and timeline; include annex in QSP • When in doubt, go back to JTAR (provisionally approved) • QSP Light approach (content more than format); need critical components • 10 days; take breaks
	Document Review	• Occ spec • JTAR (30 tasks): confirmed T/NT list
	CT0320	Discussion on use of term "parameters"= as a supervisor reporting your section and above; briefing Cert 4 replacement; originally trained at PO1 level but brought it down so they have the breadth of experience as PO2
	Task analysis: grouping	
	Main headings	Rough grouping decided to have four main headings: D/C; Work Periods; Environment, Health, and Safety; ~~Platform management~~
	Task analysis brief	Given by TDO to all group members
	Task analysis: refining scalars	
	Damage control	Narrow down to "Section base IC," doesn't include HQ1. "Coordinate" = communication side of managing; split into "to HQ1" (IT0150) and "to SBT" (IT0140). Listed all the different "stations" involved (EFS, ES, AS, BT, SSD, CBRN, Sustained fire/flood). *Tabletop instructional method recommended; scenarios at DC School; concurrent to OOD Ashore phase if possible. Similar to DCOTT scenarios.* ~~Section Base IC job has not changed. Decided by board to use existing Eng QL6 DC portion to build this training.~~
	Enviro, health, and safety (EH&S)	Lock out/tag out (UT0320): Imperative everyone gets it at a core level. Involves them to step back and see how it affects the ship as a whole; compliance with the regs. "Enforce SEMS Regs" = UT0310 and UT0320 "Brief stakeholders on EH&S reg" = AFT1010 and AFT0480
	~~Platform Management~~ Section IC	~~Broke down into "propulsion" "ancillary" "auxillary" "electrical";~~ "Section IC" = job (main task); personnel admin and management (NT because of experience / PLQ / ILQ / LRH); briefing up and down the chain (T); machinery readiness and maintenance requirements (T).

Date	Decision/Event	Detail
		CT0320 (NT; move to SWK), CT0340 (NT, Experiential), AFT0520 (Moved to maintenance); Responsibilities of Section IC = knowledge element (NT; found in ILQ). ~~*50-150 min trg*~~
	BT0080 BT0090 BT0070	~~only the outside stuff (visual), not internal inspection (internal = CPO level); while alongside. Currently no job aid checklist. All falling under platform management / propulsion~~. Will be **NT** based on TM input and board concurrence because "inspect" is lower level than a PO2.
	Work Periods	~~DRMIS Supervisor trg pre-requisite for trg~~. Front load, prior to this course? Must "evaluate" "coordinate" "compile" MLR work surveys to complete a pre-refit report (AFT0330); the evaluate, coordinate, compile works with all other work period tasks as well; trained only under work surveys.
		Board decided that tests and trials are different enough to be trained separately: tests alongside to make sure it works (equipment based), trials are under varying operational parameters. ~~"Coordinate tests" (created task)~~ "Coordinate trials" (AFT1130): ~~AFT0880 changed to be just trials with new task for tests (currently tests and trials); AFT0580 duplicated for tasks; AFT0550 changed to be just trials with new task for tests (currently tests and trials); AFT0560 changed to be just trials with new task for tests (currently tests and trials).~~
		~~UCR: elec LS start seeing this process but mech doesn't; PO2 at higher level.~~ UCR website includes a training program to demonstrate how to fill one out. Discussion as to what level PO2 are at with UCRs; given UCR user guide from http://ucrs.mil.ca/Default.asp, decided that PO2s will be OPIs and CPO/PO1 will be coordinators. "OPI is now responsible to investigate/rectify, make recommendations and report back to the Co-ordinator, via the report," (from user guide).
	Section base IC	Looking at the existing Phase from Mar ENG QL6: too in depth with stability. Not going to use. Section needs to be taught at DC school to hit on competencies (situational awareness, stress management, resource management…). Could be in conjunction with OOD trg, DCOTT, or equivalent (NTO boards?). ~~BDCS trg pre-requisite. Time needed for retests~~ Each "station" involves different products as SB IC, but the process is overall the same. "Stations" likely knowledge components / TPts. Anything "on-scene" is outside scope of the job (IT0140) = **NT**; experiential through work ups and DC trg, done by MS primarily.
	Task analysis: Review	
	PO 001 Adhere to EH&S regulations	"EO 001.01 Brief stakeholders on certification documentation" incorporates AFT1010 AFT0480: includes "Captains books" which is introduced here but reinforced throughout training as TPt/knowledge component of other EOs. *Guided discussion / interactive lecture 100 mins.*
		"EO 001.02 Enforce SEMs" incorporates UT0310 UT0320: teaching how to use and read SEMs. *3-4 pds interactive lecture for UT0310. 25-50 mins of training UT0320 (DL? Interactive lecture? have to consider SEMS accessibility on internet vs intranet).*
	PO 002 Coordinating departmental maintenance	"EO 002.01 Coordinating maintenance" incorporates AFT0530 AFT0490 AFT0520. *Interactive lectures 100 mins*
		"EO 002.02 Complete work acceptance form AFT1040" incorporates AFT0590 *Interactive lectures 100 mins*
		"EO 002.03 Coordinate MLR surveys" incorporates AFT0330. *Interactive lectures 100 mins*
	~~PO 003 Coordinate tests and trials~~	"EO 003.01 Coordinates ME tests" incorporates AFT0700, AFT0560, AFT0880, AFT1020, AFT0550, and AFT0580.

Date	Decision/Event	Detail
		Interactive lecture 200 mins
		"EO 003.02 Coordinates trials" incorporates AFT1130 and AFT1030. *Interactive lecture 200 mins*
	PO 004 Supervise Technical Administration	"EO 004.01 Action unsatisfactory condition reports AFT0720" incorporates AFT0570 and AFT0340. *Interactive lecture 200 mins*
		"EO 004.02 Action engineering changes" incorporates AFT0450. *Interactive lecture 200 mins*
	PO 005 Perform duties of SB IC	"EO 005.01 Coordinate HQ1 Support in Section Base"
		"EO 005.02 Manage section base team in action/emergencies" incorporates IT0150. *SMEs recommend 1 week (2000 mins), plus 120 mins/candidate for assessment*
	Board adjourned	
20 Feb	Reviewed ROP	
	PO writing brief	Given by TDO to all board members
	PO writing	
	"Coordinate tests and trials"	Discussion about difference between tests and trials; decided that trials are the end result therefore PO changed to "Coordinate marine engineering trials". Discussion about the size of this PO; decided that because tests and trials involve similar processes at different levels, teaching them as separate enabling objectives would be superfluous therefore decided by all board members that "Coordinate marine engineering trials AFT 1130" changed to EO under PO 002 "Coordinate departmental maintenance". Decided that training tasks involving "testing" would be redundant because trials is to a more complex level therefore AFT 1020 is a **NT**.
	"Coordinate departmental maintenance"	Discussion about PO title; decided that more appropriate PO title would be "Coordinate marine systems maintenance". Discussion about EOs; decided that "Complete work acceptance form" and "Coordinate MLR surveys" are too similar to be done separately therefore merged into one EO "Coordinate work periods".
	Priority PO 002	Board decided to make all EOs in PO 002 "Medium" because we need them to do the task competently but the impact is not high level and they will have some supervision while doing the job.
	Assistance vs Supervision	In given / denied for EOs/POs: "Assistance" is if another worker is required (or denied) to complete the work; "Supervision" is if a supervisor is required (or denied) to give guidance in completion of the task.
	References	Due to time constraints, this will be put on hold for the context of this board. We will input any obvious references as we go.
		Follow-up activity: SLt Smart will contact Lt(N) Wilson at FTA for a list of references related to tests/trials.
	Job title: Mar Tech Supervisor	Board decided the job title for the purposes of PO writing will be "Mar Tech Supervisor".
	"Supervise Tech Admin"	Board discussed validity of tech admin at the PO2 training level. Decided that UCRs and ECs should be taught at the MS/LS level since they are the ones completing the paperwork while the PO2s are just vetting the reports. Because the supervision component is experiential, it is decided that: AFT0340 - Prepare UCRs, AFT0450- Review Engineering Change Proposals, AFT0570- Review UCRs, AFT0720 – Action UCR are all **NT** at the PO2 level and should be moved to the MS or, if no RQMS is created, ~~the LS level~~ the USQs – MS (either Elec *and* Mech or combined course; already under consideration).
	Priority PO 001	Board decided to make EOs in PO 001 "High" because of the importance placed on EH&S; "if this goes badly, the CO will be hearing about it" = High impact to the unit and potentially organization.

Date	Decision/Event	Detail
	"Act as SB IC"	Board reviewed previous PO from Legacy Hull tech QL6A to see what will remain the same / what will change. They decided to cut and paste elements from the old document into the new, making amendments as required. ~~Board decided that all candidates will require valid Fit Test certification as pre-requisite for training.~~
	Priority PO 003 – Act as Section Base IC	Board decided to make EOs in PO 003 "High" because the impact is very high as it affects safety of ship and personnel.
	IT0490 "Manage SBT in action/emerg stations"	Board decided the task was NT given the similarity with IT0150 (Coordinate SBT). They decided "coordinate" was more in line with what a PO2 would be doing, keeping that task in lieu of the "managing" piece.
	Stakeholder discussion	From LCdr Day: would like to see ~~DRMIS supervisor training~~ on the RQPO2 course. ~~Board decided they would like to see it added as the primary module on the course, with the remaining RQPO2 being the second module.~~ This will allow for those who already have the MITE to be loaded on the second module without having to redo the DRMIS supervisor course and will allow other CAF mbrs to be loaded on the first module to meet zero load numbers.
		From LCdr Day: would like to see firefighting and flood refresher added to SBT IC portion of the course. Board decided that the refreshers are **NT** based on the experience of the target audience: candidates do not need current knowledge of how to do firefighting/flood before learning the SBT IC position and they would all have several refreshers under their belt by the PO2 level making the refresher redundant.
	New PO numbering	Board decided that with the DRMIS Supervisor component becoming a separate module required for successful completion of the remainder of the course, the DRMIS Supervisor course would become PO 001. Due to its importance and relative difficulty when compared to the rest of the material, "Act as Section Base I/C" would become PO 002. "Adhere to Environmental Health and Safety Regulations" would become PO 003, and "Coordinate Marine Systems Maintenance" would become PO 004.
	Annex D completion	Board completed Annex D, capturing the complete scalar for posterity.
	TEL discussion	Board discussed TEL requirements. Based on SME expertise, Board decided that the procurement and use of a BDCS trainer, independent of the coastal DCTFs, would be an ideal aid to train candidates as Section Base I/C. Discussion captured at Annex L.
	Board adjourned	Follow-up activity: SMEs to brainstorm partial list of useful references for later capture.
21 Feb	Reviewed ROP	Board chair input on TEL discussion: newly procured BDCS Trainer should belong to MS Div at each NFS (to supplement BDCS trainer found in each DCTF)
	Target Population Analysis (TPA)	
	Brief	TDO gave brief on TPA to entire board
	Experience	Discussion about the experience level at sea and in rank prior to being given the course; board decided that it should be for "substantive MS" (i.e. PLQ/equivalent), merit-loaded based on board ranking/PERs. Board proposed requirement for ~~6 months at sea as a MS~~ prior to being eligible for the training.
	Age	Varied; personnel are joining older therefore range from early-twenties to late-forties
	Competencies	"Stress-management" etc: captured in PER / merit ranking therefore not specifically addressed here
	Learning skills/preferences	Situational / scenarios; context is important.

Date	Decision/Event	Detail
		Access to computers important; transition from "putting down the tools" to administrative responsibilities.
	Networking	Knowing contacts; putting faces to names of key players
	Design: PC Planning and Checklists	
	Brief	Given by TDO to all SMEs
	"Adhere to EH&S Regulations"	Discussion on best method of assessing: written exam? DL portion? Scenario? They want higher level then knowledge component: they'll have to use their judgement. Board decided good option is In-basket exercise with multiple scenarios and various inputs (of both high and low level of difficulty) to add context and stress administrative requirements versus hands-on to hit on the higher-level / supervisory component. SMEs have many real-life scenarios that could work if developers require examples.
		Board decided that main performance criteria are prioritization of tasks and the content of the (informal) after action report, which will justification of their decisions and references used. Follow on discussion determined that document itself would be appropriate for assessing prioritization
	"Coordinate Marine Systems Maintenance"	Board discussed best method of assessing this performance: scenario where they are given a 10 week message and have to detail follow-on actions.
		Discussion as to scope of the different work periods: DWP infrequent but important (medium sized process), MLR once per career (possibly) and huge process, general maintenance small process and frequent. Scenario will be based on a TRP as this will help them see the importance of looking ahead in the schedule (commonly seen on the job).
		SMEs decided that the scenarios will have to be mech/elec specific. Though the process (i.e. critical element) is the same, the results will be a bit different and having them deal with familiar equipment will be preferred.
		SMEs decided that performance criteria will be based on process for lead in work, process for discussion ("bun fight" justifying their prioritization), product in "8 week" message.
	"Act as Section Base IC"	Board reviewed existing PCs and decided that the existing PCs are not sufficient for the Mar Tech RQPO2 purposes. Creating PC details from "scratch".
		Board decided that communications are critical success criteria because of the potential impact to the unit safety as well as command and control (imperative that HQ1 is aware of what is happening with SBT).
	Naval Technical Communications course	
	Module C	Existing QSP; Module C is currently geared toward QL6/QL6A level MSE trades. Board has decided (no chair) that this module will be taught with the Mar Tech RQPO2 as written in the existing QSP. Mod C is 1725 training minutes (not all concurrent); Ch 1 and annexes will need to include timings and scheduling for Nav Tech Comm Mod C. Legacy personnel may already have this training; as such, an independent module may be the best means for delivery.
	Board adjourned	Board adjourned at 1445 due to heavy snowfall impact on travel.
22 Feb	**Design: PC Checklist**	
	"Act as Section Base IC"	Board decided they want to confirm applicability of existing checklist therefore contacted NFS(P) DC Div to get it sent over. SMEs believe that the checklist will be sufficient with "no need to reinvent the wheel".
		Discussions about verbs used for checklist; emphasis on performance-oriented, keeping it observable and measurable.
		Board decided to create their own checklist using the QL6B DCCPO and QL6A SBT I/C existing checklists as guides to ensure nothing was forgotten. Some checklists points were based on behavioural indicators / competencies that are required for the SB I/C.

Date	Decision/Event	Detail
	Design: Writing EOs	
	Brief	Given by TDO to SMEs
	EO 002.01 "Coordinate section base team training"	Discussion about whether or not to use the Hull tech QSP as a skeleton for the EO; SMEs decided that it would be just as fast and more inclusive to create from scratch (existing QSP can be used as a review afterward as required to check that nothing was missed).
		Discussion raised as to the evolving job titles after amalgamation meaning that some terms may change. For the purpose of this QSP: • DCPO (Sr legacy hull tech) = Sr mechanical manager • CEng = Departmental coordinator • SnrET = Snr electrical manager
		Board defined "Requirements" in the context of DC elements to include personnel, equipment, and tasks
	EO 002.02 "Brief SBT members on DC considerations"	Board hashed out a suggested EC guideline as well as discussing EO guidance including activities and timing. Decided that practice would be important, but the overall scope is generally fairly small leading to only a small amount of time allocated for the actual EO.
	EO 002.03 "Coordinate a SBT in actions/emerg stations"	Discussion as to what knowledge elements are involved and decided that knowledge components were from previous EOs with this one being more of a consolidation of .01 and .02 and a practice for the PC. Most of the EO will be practical activities.
	New BDCS Trainer (TEL option)	NFS(P) MS Div have basically the entire configuration from DC school except for LSDs (Large Screen Display; touch screen). The other part which would improve the training and save time because of multiple assessments / practices happening concurrently would be to get a software patch which would allow the existing network to actually run with the same scenario (as it stands right now, each console is individual which means less realism with scenarios). The East Coast supposedly has the exact same thing (not confirmed, but that's what PO2 Jack Sleeman, the IPMS/MCRT/PCOTT POC at NFS(P) said). Impact of status quo on PO2 MAR TECH training moving forward was discussed; SMEs determined that training could still be conducted, though it would lack the realism of having multiple 'Section Bases' operating in concert.
23 Feb	EC 002.01 "Coordinate SBT training"	Discussion for how to assess the EO. Board chair indicated concern that time was being misallocated by "reinventing the wheel" with SBT IC training however board decided that the existing training focused too much on theory and not enough on practice therefore rebuilding was preferred since the time to do so is available. Board decided on a practical assessment having candidates prepare a plan for SBT trg to better prepare the trainees for on the job performance.
	Design: Writing EOs	
	EO 002.01 "Coordinate SBT training"	Board discussed activities to present the theory component of the lesson and decided a guest SME lecturer with practical/real-life experience would be ideal to present some of the material in more detail.
	EO 003.01 ~~"Briefing stakeholders on cert docs"~~ "Managing Captain's Ship's Books"	Discussion about task AFT1010 verb "Complete"; board decided "Organize" is more appropriate as they will be "arranging into a structured order / coordinating" the documents (used rental car example: someone needs to allocate cars based on maintenance routines, etc which is equivalent to what is being done with captain's books)
		Board decided "Manage" is a better main task than "brief" as briefing is part of the management aspect. "Manage" was chosen based on the JTAR definition "To direct, govern, or control in action or use". Board decided that "briefing" is previously taught therefore NT.

Date	Decision/Event	Detail
		Board decided that a SME lecturer from FTA would be best to explain use of Captain's Ship's Books because they are the most familiar with the document requirements.
	EO 003.02 "Enforce SEMS"	Board decided that the EC would be written but scenario based where they are still gathering information practically from the refs to properly troubleshoot the given issues. This should still adequately prepare them for the practical PC.
		EC: Board decided trainees can have access to the intra/internet and use any means (references) to get the relevant information because the information is so critical (safety).
		Board decided SME lecture from Safety Officer / Formation safety would be appropriate to discuss in general about SEMS.
	"Safety Management Course" and "Unit Environmental Officers Course"	Board discussed validity of these two courses in context of RQPO2. It was decided that they will likely form part of the Mar Tech PO2 performance on the job and are therefore potential additions to the course. Board Chair to look into the applicability of the courses and whether they would be beneficial EOs for training.
		LCdr Day input: He believes that the requirement is already covered because the MS and CS EOs are required to have this training therefore PO2s should not be required to have it.
		Board still feels as though safety is not stressed enough in training and TDO is currently investigating a possible safety training gap in legacy engineering training, as well as the MAR TECH RQ-OS course. Moving forward, this gap in Mar Tech could potentially be filled by delivering this existing training. Formalized safety training isn't really found within Legacy courses; as such, moving forward, it would be beneficial to the occupation to receive dedicated safety training.
	EO 004.01 "Coordinate departmental maintenance"	Board decided a SME lecture (Ship's Support Officer) will be appropriate because they will be best to give context.
		"Complete pre-refit survey report" task added as a TPt from the "work periods" EO which was decided to be NT.
	"Coordinate work periods"	Board decided that this task was **NT** because the process is the same as coordinating departmental maintenance.
	EO 004.02 "Coordinate Marine Engineering trials"	Board decided a series of guest lecturers would be appropriate because each one will have a particular area of expertise that they will speak to.
		For the second lesson objective, the Board examined the aim of conducting follow-on activities. However, this discussion is to be concluded Monday morning.
26 Feb	Reviewed RoP	Possible case study (EH&S): HMCS Calgary fuel spill that happened over the weekend (30000L)
	Board chair	Board chair not present for the entire day.
	Modularization	Board decided that there would not be any modules for the RQPO2 because the course was small enough in scope that modularization wasn't necessary.
	EO 004.02 "Coordinate Marine Engineering Trials"	Amended EC to include questions on risk assessment and waivers. Written EC is sufficient because of the material covered
		SMEs decided that a couple case studies are the best method of instruction following a short lecture so that the trainees have the chance to see some real-world examples, discuss how they would handle it in small groups, then find out what the COs actually decided so they can see the similarities and differences.
	"Safety Management Course" or "Supervisor Safety	Discussion with Stakeholder: SMEs gave their reasoning for why they believe these courses should be incorporated into the trg: • Mar tech PO2 will have direct input on how safety is handled even if they don't directly brief command

Date	Decision/Event	Detail
	Course" and "Unit Environmental Officers Course"	• Currently there is little to no training regarding environmental and safety throughout the Mar tech courses Discussed that there are currently no tasks that justify inclusion of extra safety training.
		Decided by stakeholder and SMEs that UEnvO course is not applicable at the PO2 level at this time; discussed that this creates a gap at a level lower than UEnvO therefore it should be investigated whether a lower level Environmental course should be created.
		Board decided to include as many tasks/TPts as possible to cover off the environmental aspect for the current board knowing this may be changed if a formal environmental trg program is created.
		Decided by stakeholder and SMEs that SSC is applicable at the PO2 level (unless / until a Mar Tech specific Safety / environmental training program is implemented). .5 days of existing training given through MARPAC Formation Safety (is not listed as trg for MARLANT).
		Decided by stakeholder and SMEs that SMC course is likely more applicable at the PO1 level and should be re-evaluated at the RQPO1 board. Stakeholder to discuss with TM to incorporate
	BDCS trainer update at NFS MS Divs	CNPTG has given go ahead for trainer. Board to initiate LSR / FA for acquiring required components: • LSDs (touchscreen) x4 • Software patch to allow consoles to reflect each other • IPMS console
	CTAT	For developers: there are some TPts which will deal with controlled substances where CTAT considerations will come into play (eg. Dealing with Sonar fluids) therefore these phases may have to be developed in house instead of off-site. Any that were identified during the board are highlighted in red throughout Ch 3.
	EO 003.02 "Enforce SEMS"	Board decided EO divided into three main LOs: "Supervise Environmental", "Supervise Safety", and "Supervise LOTO" s this accurately reflects the three main aspects as seen on the job.
		Board discussed ~~potential DL component.~~ Because much of the information is theory based, could be self-study instead of instructor led.
	JTAR skill/knowledge review	Board reviewed the skill/knowledge assigned to Trained tasks at the RQPO2 level. • "Assessing systems" skill: linked to "analysis" • "Interpreting" skills: linked to the case studies / scenarios throughout instructional activities • "Technical Communications" skill and knowledge: to be covered off by the existing Mod C course
		Amend JTAR: AFT0490 included skill "Planning activities" at level 3
	Ch. 1 review	
	Personnel Resources	Board decided to put Mar Tech requirements with the legacy requirements in brackets to be removed when no longer applicable.
		Board decided the Sr instructor = PO1 (manager) based on NFS construct.
		Board decided to include two instructors per course, one elec (with USQ and sea tour), one mech (with USQ and sea tour) so each can speak to their specialty. Two instructors is mandatory for any size course for safety reasons in the SBT IC phase of training.
	Course capacity	Board will contact Occupation manager to find out estimated number of personnel requiring the RQPO2 in a given year to better judge course capacities.

Date	Decision/Event	Detail
		Board decided on a maximum load of **12** because of DCTF capacities. Any more and it would be extra time added for individual assessments. 12 is also the current standard for most legacy training programs.
		Optimum of 10 based on number of spots in the MS Div BDCS trainers at the school.
		Board recommended 8 as a best guess, feeling anything less would be a waste of resources. However, based on break down of actual minimums to run the course, a minimum of 3 was decided as this would still allow for SBT IC training as well enough input for debates / discussions / case studies, etc.
	Pre-requisites	~~SMEs decided that SWK qual gives them breadth of experience required to be an elec or mech supervisor therefore should be a pre-requisite. To be confirmed by TM/Occ mngr.~~
		SMEs decided not to consider submariners in pre-reqs therefore they may need special consideration. ~~To be confirmed by TM/Occ mngr.~~
	Tiller Flats (parking lot) review	
	Tech Comms Mod C (118130)	SMEs decided that the module will be incorporated based on the existing QSP as PO 005 in the RQPO2.
	Conditions for EOs	Was put in parking lot in case there was no time to fill in; board has the time and filled in conditions as required.
	EC Checklists	Was put in parking lot in case there was no time to complete; board has the time and filled in EC checklists for every EC (other than those found in existing QSPs).
		"Enforce SEMS" EC 003.01: candidates to answer questions (written) based on a scenario. No need for a checklist (not task-based) but board created possible exam questions to demonstrate the desired assessment outcomes.
27 Feb	EC 004.01 "Coordinate maintenance"	~~Board discussed validity of DLN assessment; no decision made.~~ Created questions etc based on possible scenarios.
	PC 002 "Act as Section Base IC"	Board discussed "retest time"; it was decided that instead of booking specific time for retests, time would be amended from 60 to 75 minutes for each candidate to ensure that the resource is booked for sufficient time. This time may change after pilot course depending on scenarios and success rate.
	PC 003 "Adhere to EH&S regulations"	Board discussed timings: decided on 200 minutes for each section (vetting and prioritization, prep, completing final document) = 600 minutes total with all candidates being assessed at once.
	EC/PC Pass / Fail criteria	
	EC 002.01 "SBT Trg"	Discussed critical items in the checklist. Reviewed points to ensure they were observable and measurable. Decided on the 70% mark (7/10 hitting all criticals) because this hits on safety (more than 60%) but isn't an emergency (less than 80%).
	EC 002.02 "Brief SBT on DC considerations"	Discussed making all items critical because of safety component. Was decided there are a couple items that are not safety related therefore there will be some non-criticals. Decided on 80% mark because it could be life/death (emergency). Discussed making it a Critical EC but it was decided that it's unlikely to have failures and it is still a formative assessment therefore remedial training and retesting is sufficient impact.
	EC 002.03 "Coordinate SBT in actions and emergencies"	Board decided that all items will be critical because of the high impact on life associated with the role of SBT IC.
	Percentages for ECs and PCs	If an EC or PC has a 60% pass mark, it is because it does not have direct and immediate safety outcome.

Date	Decision/Event	Detail
		If an EC or PC has a 70% pass mark, it is because it has a safety outcome but doesn't directly impact life/death situations. If an EC or PC has an 80% pass mark, it is because it is for an immediate safety outcome. If an EC or PC requires all critical items, it is because it has a high life/death safety outcome and any failure could result in death on the job.
	QSP Review and Confirmations	
	SB I/C Video	Board decided that based on scope of lesson, the instructor might as well deliver a standardized brief to the class as an example to conserve development resources. If a larger encompassing video is created for HQ1 for the RQPO1, the section base component could be used at that point as the "perfect example" for SBT IC training.
	QSP Formatting	The Board decided a review of the document to ensure proper vocabulary and formatting would be a good use of time. Additionally, rather than copy training from DRMIS Course, Supervisor's Safety Course, and Naval Technical Communications Mod C, that guidance be provided to use these existing courses to satisfy training requirements. This way, should one of these elements change, a review of this QSP will not be required.
	DL	Given the small amount of information that would be on a DL course and the limited benefit (if any) involved, the board decided against having a DL component. The importance of discussions and learning from peers was also an element leading to the decision against DL.
	Course Capacity	TM agreed that 12, 10, and 3 (Max, optimal, min) makes sense. He confirmed there would be roughly 3 courses per year per coast.
	Safety Supervisor Course	Follow-up required to ensure that the formations are ok with delivering this training (encouraged by TM). TM is concerned there might be a problem incorporating Formation Safety (contracting, cost, etc). TDO to reach out to FSE.
	DRIMIS Supervisor	TM concurs that everyone will be required to do the DRMIS supervisor course. If they already have it, it will be a good refresher and those who are proficient can help their peers.
	Tech Comms	Board discussed having some of the requirements for tech comms completed prior to course start. Board decided that it will be written in the joining instructions for each candidate to come on course with: • Technical topic already approved by instructor • Collection of references (initial research) There will be a database maintained by the NFS Mar Tech Div Chief with a list of topics already covered so that there are no repeats.
28 Feb	**RoP Review**	
	BDCS Training as pre-requisite	Everybody will have it; can't make it to the senior MS level without it. Could be listed as pre-requisite but not required.
	Fit test as pre-requisite	If TEL approved (upgrade to BDCS Trainer), then no requirement for a valid Fit Test as a pre-requisite because they won't be at DCTF. If they are required to go to DCTF, they will need a valid Fit test.
	BDCS Trainer Upgrade	LCMM (LCdr Mascarenhas) does not have any existing components therefore procurement will be required for trainer upgrades.
	TRP Review from FTA	Tiered Readiness Program – MARPAC Trials Procedure should be a reference for trials PO/EO. Added as a reference.
		An "example" / historic trials schedule (example from CPO2 Wilcox) to be used as a training Aid for EO 004.02 "Coordinate Marine engineering Trials"

Date	Decision/Event	Detail
	SWOHS course / SSC	Supervisor Workplace Occupational Health and Safety (SWOHS) is being developed by D Safe G. This would likely replace the current MARPAC Safety Supervisor Course.
		MARPAC Formation Safety is ready to support SSC or SWOHS training for the Mar tech RQPO2. Not yet confirmed from MARLANT Formation Safety
		Board reviewed the draft TP for SWOHS course for applicability. SMEs decided that it incorporates everything that they would need at the PO2 level (on the safety side).
	SWK Pre-requisite	From TM: SWK cannot be a pre-req for the course because some personnel from different platforms will not have enough (any?) time on a CPF and SWK is CPF specific training.
	Submariners	The RQ PO2 training will be for submariners as well. Board discussed that they will likely have a steep learning curve.
	QSP Review	
	Pre-requisites	It was decided that PLQ and rank of MS were still valid pre-requisites. The SMEs decided that it was also important for each candidate to have already completed the Mech or Elec MS USQ (this reflects the "Mar Tech Trade Progression Diagram v1.3).
	Full review	Board reviewed the final document, prior to submission to the stakeholder
	Adjournment and Follow-On	
	Adjournment	Board adjourned. TDO to recall board members as required to conduct follow-on activities.
	Board Chair meeting	TDO and Board Chair met to capture Board Objective for RoP and to discuss Board process.
	Lessons Learned	SMEs solicited for input into Lessons Learned (following)
10 May 18	RQ PO2 Mar Tech Supr QSP update	Upon completion of the USQ PO2 Mech/Elec QSPWBs, it was determined that training to manage personnel (from both a divisional and readiness level) was missing. This was presented to D Nav P&T and support was given to include this training. To effect these changes to the RQ PO2, the following personnel came together from 7 – 11 May 18: Lt(N) Robinson, TDO, NTDC(P); Lt Tzotzos, TDO, NTDC(P); CPO2 Zsombor, Mar Tech Dept Coord; and PO1 Penney, Mar Tech Elec Mgr (NTDC(P). The following changes will be made to the QSP (driven by the Elec Supr QSPWB): 1. Addition of PO – Manage Personnel Requirements. The email below from CPO1 Pryor, D Nav P&T, reviewed tasks at both RQ PO1 and RQ PO2 and made the recommendations below. It is important to note that these tasks were reviewed and final disposition of tasks below can be found at Anx B of this QSP. **From:** Pryor CPO 1 BA@C Navy D Nav P&T@Ottawa-Hull **Sent:** May-03-18 8:33 AM **To:** Tzotzos Lt LA@NPTG NTDCP@Esquimalt <LIZA.TZOTZOS@forces.gc.ca> **Cc:** Desormeaux LCdr C@C Navy D Nav P&T@Ottawa-Hull

Date	Decision/Event	Detail
		<Chantal.Desormeaux@forces.gc.ca>; Gagné CPO 2 YS@NPTG HQ@Esquimalt <Yannick.Gagne2@forces.gc.ca> **Subject:** FW: Stability - Final Answer Request Hi Ma'am, So, I looked at the JTAR and the various notes and gave everything some deep consideration here is the direction regarding the PO1 RQ: PO 002 Will Remain in the PO1 RQ PO 003 (Docking and Undocking) will Remain in the PO1 RQ in its entirety. PO 004 EO 004.01-Act as Electrical Manager for Emergency/Action Stations is to be moved to the Elect Supervisor USQ. PO 004 EO 004.02 - Act As DCPO is to be moved to the Mech Supervisor USQ. PO 004 EO 004.03 will be common to both of the PO2 USQs. This doesn't mean that it is to be moved to the PO2 RQ but needs to be present in both the Mech and Elect Supervisor USQs. Note The Role of the DCPO/ DCO on the MCDVs will be covered off by a separate piece of training (TBD if required). PO 005 Contains a couple of things which are more expansive than indicated in the QSP Managing Risk Assessment (AFT0170) – this needs to be broken out and taught as part of a larger Naval Materiel Assurance topic Managing short work periods (AFT0210) - This is one of the key tasks that PO1s and CPO2s will carry out. As well there are a few Tasks which are not covered at all: These are not learned properly through experience alone and need to be included at the PO1 level to prepare the member their role as CHOD. Note: Some of these topics are also part of the WEng training for the CPO2. I attach it as an example. AET1010 – Coordinate training and readiness – Both Personnel and Materiel (ie CRRs)/ captured at CPO2 Level needs to move to PO1 Common AFT0080 – Manage marine engineering trials agenda – Readiness / appears at PO1 Maint Manager, Needs to be added to PO1 Common AFT0090 – Manage maritime evaluation activities- appears at RSB Ops coord Needs to be added to PO1 Common AFT0100 – Manage material certification program – This is also in Naval Materiel Assurance/ Appears at PO1 Ship Sub Support Mgr Needs to be PO1 Common AFT0820 – Advise on particularized maintenance & repair schedule (PMRS) – Refits/ dockings/ NT in Elect Mgr. Needs to be PO1 Common

Date	Decision/Event	Detail
		AFT0840 — Advise on ship's standard maintenance & repair schedule (SSMRS)- Refits/ dockings/ is NT , should be T AKT1200 — Manage E&M inspections Readiness / is NT , should be T AKT1240 — Manage operational deficiency report requirement Readiness / is NT , should be T AT0180 — Coordinate equipment and system inspection Readiness/ is NT , should be T AT0190 — Coordinate systems trials Readiness / is NT , should be T ZT0070 — Manage notice for power Readiness / NT at Mech Mgr should be T PO1 Common Finally; The stability in the PO1 RQ serves the DCPO function as well as the CPO2 Dept Coord function. The level that exists in the PO2 RQ will feed into both the PO2 Mech Supervisor training and the PO1 RQ nicely. I do not feel that the calculations or concepts will be troublesome for the Electrical Sub Occ as they will be accustomed to much more complicated mathematics.

Lessons Learned:

1. Small board was very effective; however, it becomes even more important that the best personnel available are tasked, as was in this case.
2. Review RoP every day to confirm decisions and avoid duplication of training. Additionally, it can be valuable to go over milestones achieved along with the RoP in order to provide context to Board members.
3. Board composition. SMEs indicated that a Board Chair that is more available would be valuable.
4. Writing PCs at the beginning of the Design Phase was valuable, as it kept the Board focused on job performance. From there, as was seen during the Board, beginning with EC assessment plans (and checklists) helps keeps that element of performance in mind as Lesson Objectives are determined. From there, Teaching Points and then Conditions and the Standard follow naturally.
5. QSP template. The given QSP template should be properly formatted for the easy input of data.
6. Assessment criteria. Stress the importance on the student understanding the criteria instead of the instructor so that they give more detail.

Guidance to Instructional Staff:

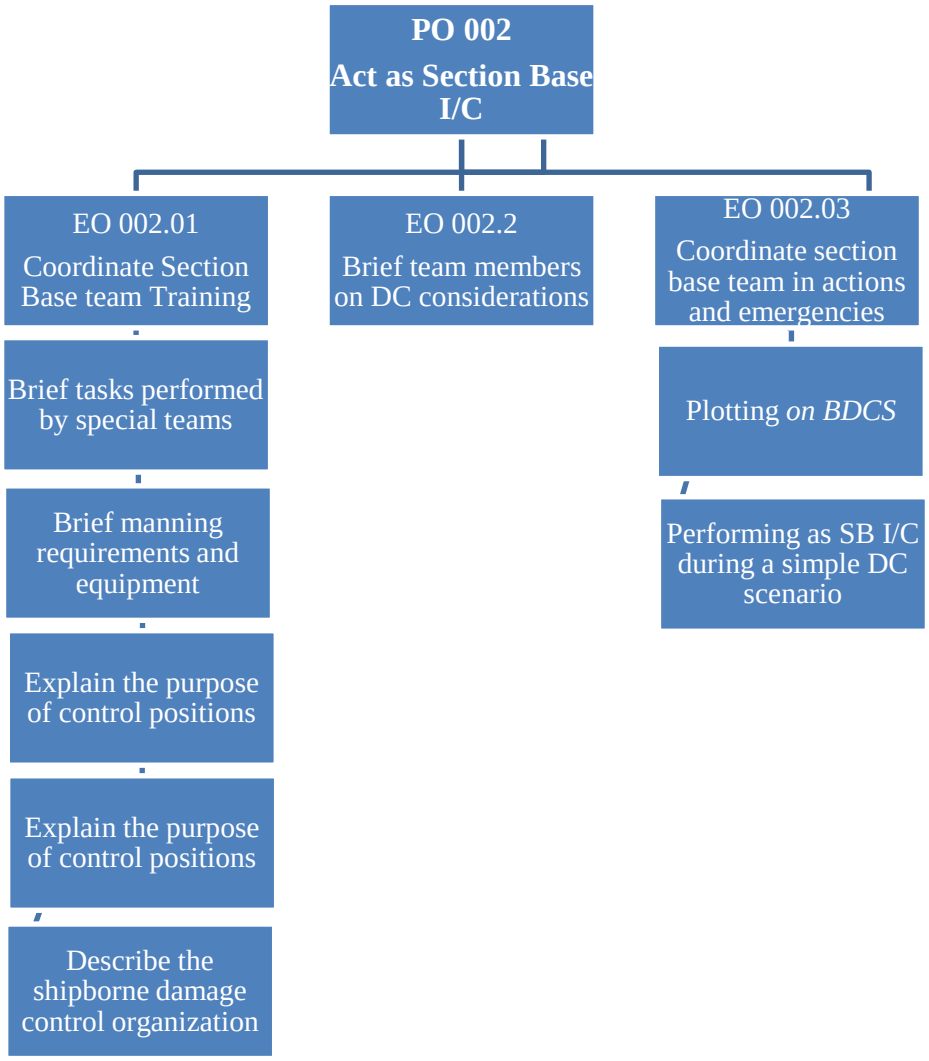
Follow Up Action:

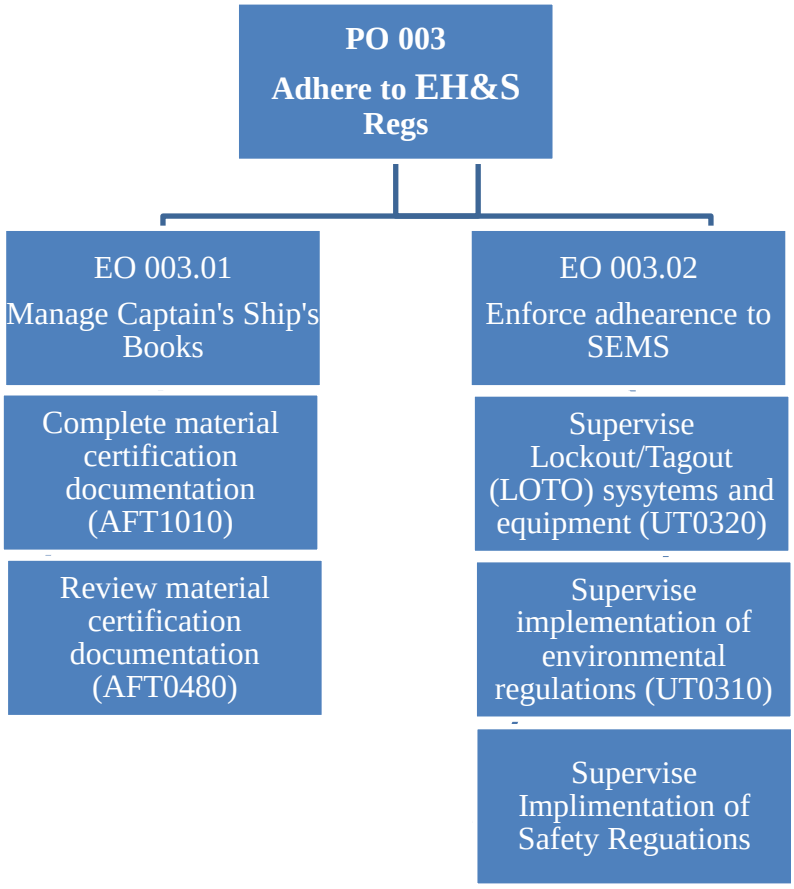
Item	Action	OPI	Expected Date of Completion
1	SPQR to update the task list with relevant tasks	TM to send to DPGR	
2	Upgrade of coastal BDCS Trainers	NTDC(P)	
3	Confirmation of Supervisor Safety Course/planned replacement on both coasts for Jan 19	TDO/FSE	
4	Courseware to be developed by contractor		31 Dec 19

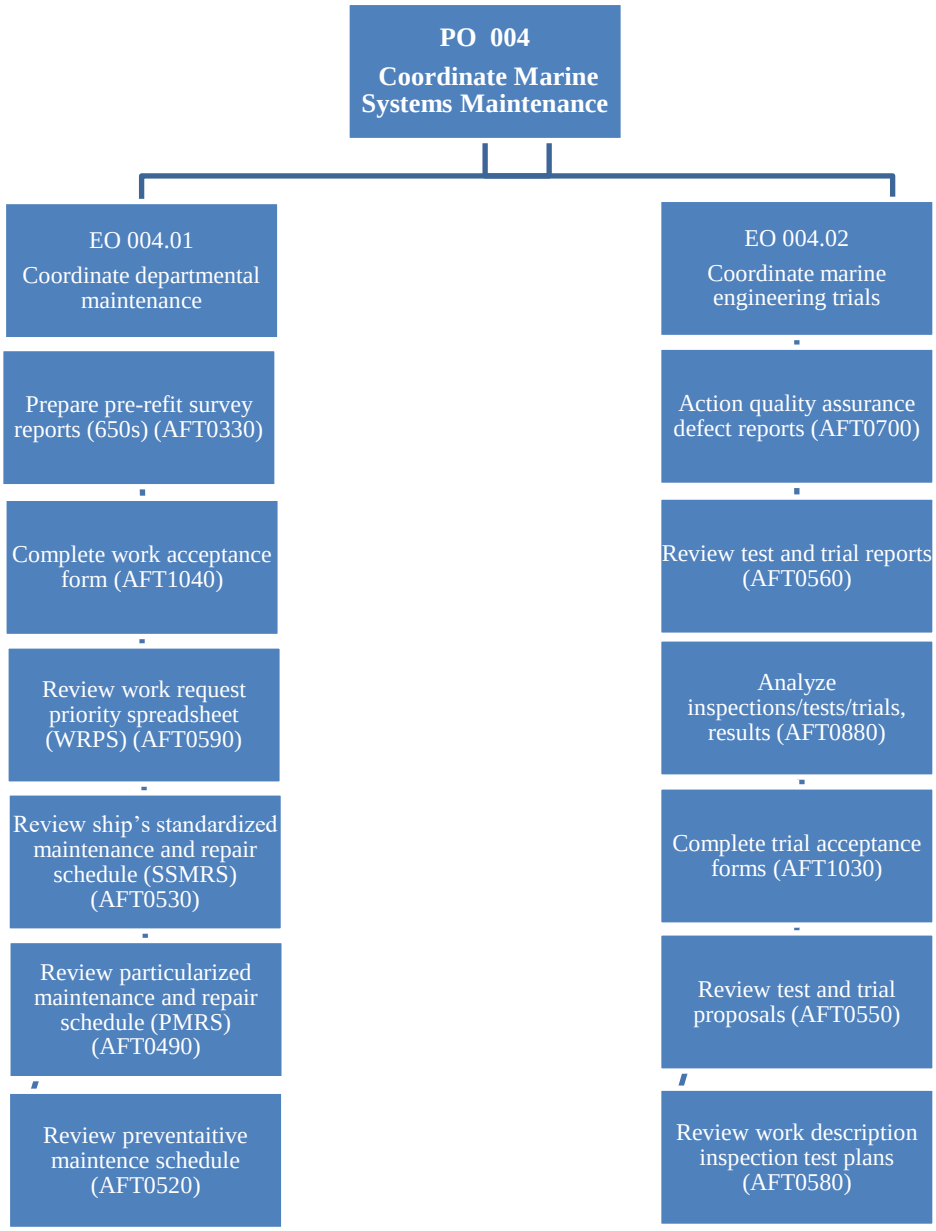
Other Considerations: Nil.

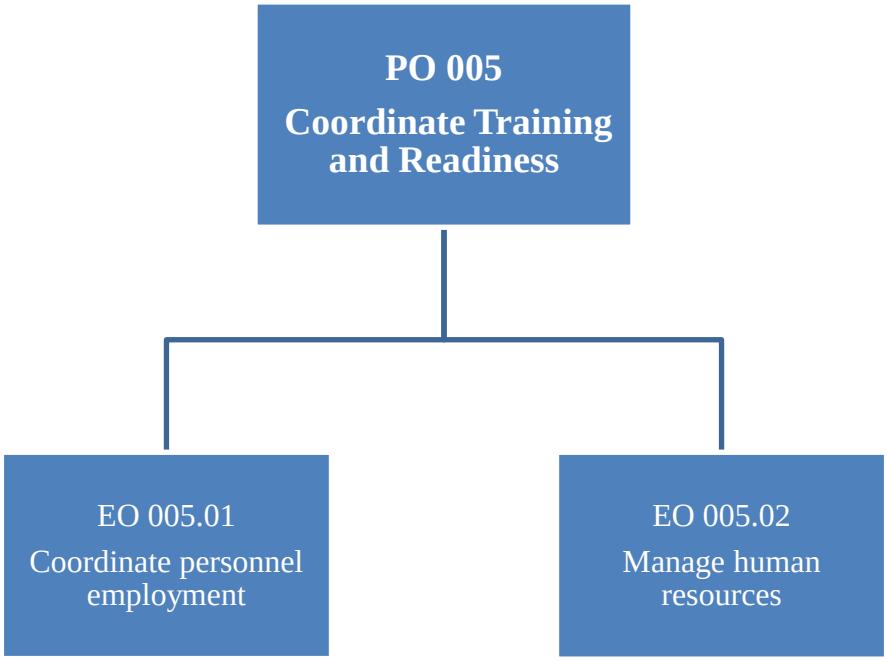
ANNEX D – **QSPWB SCALAR**

PO 001 is captured by the Naval Technical Communication – Module C, and will be a standalone module. The MITE code for Technical Communication – Module C will be awarded on the completion of RQ PO2.




PO 003
Adhere to EH&S Regs
EO 003.01
Manage Captain's Ship's Books
Complete material certification documentation (AFT1010)
Review material certification documentation (AFT0480)
EO 003.02
Enforce adhearence to SEMS
Supervise Lockout/Tagout (LOTO) sysytems and equipment (UT0320)
Supervise implementation of environmental regulations (UT0310)
Supervise Implimentation of Safety Reguations


PO 004
Coordinate Marine Systems Maintenance
EO 004.01
Coordinate departmental maintenance
Prepare pre-refit survey reports (650s) (AFT0330)
Complete work acceptance form (AFT1040)
Review work request priority spreadsheet (WRPS) (AFT0590)
Review ship's standardized maintenance and repair schedule (SSMRS) (AFT0530)
Review particularized maintenance and repair schedule (PMRS) (AFT0490)
Review preventaitive maintence schedule (AFT0520)
EO 004.02
Coordinate marine engineering trials
Action quality assurance defect reports (AFT0700)
Review test and trial reports (AFT0560)
Analyze inspections/tests/trials, results (AFT0880)
Complete trial acceptance forms (AFT1030)
Review test and trial proposals (AFT0550)
Review work description inspection test plans (AFT0580)


PO 005
Coordinate Training and Readiness
EO 005.01
Coordinate personnel employment
EO 005.02
Manage human resources

PO 006 is captured by DRMIS Maintenance Manager - 119098**,** and will be a standalone module. The MITE code for DRMIS Manager will be awarded on completion of RQ PO2.

ANNEX E – **ON JOB PERFORMANCE RECORD (OJPR)**

There will be no OJPR for the MAR TECH RQ-PO2 course.

ANNEX F – **ACRONYMS AND ABBREVIATIONS**

Acronym and Abbreviation	Term or Definition
AP	Attach Post
BDCS	Battle Damage control System
CBRN	Chemical Biological Radiological Nuclear
CFCD	Canadian Forces Controlled Document
CFTPO	Canadian Forces Training Planning Order
DC	Damage Control
DCPO	Damage Control Petty Officer
DCTF	Damage Control Training Facility
DRMIS	Defence Research Material Information System
DWAN	Defence Wide Area Network
DWP	Docking work period
EH&S	Environment, health, and safety
ERT	Emergency Response team
FA	First Aid
HQ1	Damage Control Head Quarters
HVAC	Heat and Ventilation System
IC	In charge
IR	Imposed Restriction
JPSU	Joint Personnel Support Unit
LOTO	Lock-out / tag-out
MARPACORD	MARPAC Orders
MELs	Medical Employment Limitations
MLR	Mid-life refit
NaMMS	Naval Maintenance Management System
NAVORD	Naval Orders
NEM	Naval Environmental Manual
NEMTI	Naval Engineering Manual - Technical Instructions
NW	Code to reflect work outside the scope of ship's staff; often specialty requirements that require specialized equipment (large-scale NYs)
NY	Code to reflect work usually conducted by ship's staff
PCC	Personnel Coordination Centre
PEL	Periodic Engineering Letter
PG&D	Power Generation & Distribution
PMRS	Preventative maintenance and repair schedule
RAT	Rapid Attack Team
Regs	Regulations
SBT	Section Base Team
SEMS	Safety and Environmental Maintenance System
SME	Subject matter expert
STG	Sea Training Guide
SWK	Senior Watchkeeper

TD	Temporary Duty
TRP	Tiered Readiness Program
WRPS	Work request priority spreadsheet
2I/C	Second in charge

ANNEX G – ***SUGGESTED SEQUENCING AND TIMINGS***

EO ###.##	EO Performance Statement	Personnel resources	Required Resources (for scheduling)	Trg Time per resource (mins)	EC Time (mins)	PC Time (mins)
PO 001: Draft Technical Serve Paper				Total PO time:		1825
PO ATT:	1825	PO Admin Time:		PO PT time:		
EO 001.01	Produce a Technical Service Paper	Instructor	classroom with DWAN stations and projector	550	500	
EO 001.02	Present a Technical Information Brief	Instructor	classroom with DWAN stations and projector	200	200	375
PO 002: Act as Section Base I/C				Total PO time:		3750
PO ATT:	3750	PO Admin Time:		PO PT time:		
EO 002.01	Coordinate section base team training	Guest Lecturer (DC SME from ST(P/A)	classroom with DWAN stations and projector	200		
EO 002.01	Coordinate section base team training	Instructor	classroom with DWAN stations and projector	50	300	
EO 002.02	Brief team members on DC considerations	Instructor	classroom with DWAN stations and projector	100	250	
EO 002.03	Coordinate a SB team in actions and emergencies	Instructor	BDCS Trainer	1800	300	750
PO 003: Adhere to Environmental, Health and Safety Regulations				Total PO time:		1150
PO ATT:	1050	PO Admin Time:		PO PT time:		
EO 003.01	Manage captain's ship's book	Guest Lecturer (FTA)	classroom with DWAN stations and projector	100		
EO 003.01	Manage captain's ship's book	Guest Lecturer (MS ENG post HOD)	classroom with DWAN stations and projector	50		
EO 003.01	Manage captain's ship's book	Instructor	classroom with DWAN stations and projector	100	200	
EO 003.02	Enforce adherence to SEMS	Guest Lecturer (SEMS SME, like Formation Environmental O)	classroom with DWAN stations and projector	100		
EO 003.02	Enforce adherence to SEMS	Instructor	classroom with DWAN stations and projector	100		

EO 003.02	Enforce adherence to SEMS	Instructor	RCN Vessel	100	200	200
PO 004: Coordinate Marine Systems Maintenance				Total PO time:		1600
PO ATT:	1600	PO Admin Time:		PO PT time:		
EO 004.01	Coordinate departmental maintenance	Guest Lecturers: SSO and N37 Rep	classroom with DWAN stations and projector	100		
EO 004.01	Coordinate departmental maintenance	Instructor	classroom with DWAN stations and projector	600	100	
EO 004.02	Coordinate marine engineering trials	Guest Lecturers: FMF Trial Officer or Chief, FTA SME, N37 Rep	classroom with DWAN stations and projector	100		
EO 004.02	Coordinate marine engineering trials	Instructor	classroom with DWAN stations and projector	200	100	400
PO 005: Coordinate Training and Readiness				Total PO time:		1200
PO ATT:	1200	PO Admin Time:		PO PT time:		
EO 006.01	Manage human resources	Monitor MASS Instructor	classroom with DWAN stations with Monitor MASS, and projector	400		
EO 005.02	Coordinate personnel employment	Guest Lecturer - PCC	classroom with DWAN stations and projector	100		
EO 005.02	Coordinate personnel employment	Instructor	classroom with DWAN stations and projector	100		
EO 005.02	Coordinate personnel employment	Guest Lecturer - JPSU	classroom with DWAN stations and projector	100		
EO 005.02	Coordinate personnel employment	Instructor	classroom with DWAN stations and projector	200		
EO 005.02	Coordinate personnel employment	Instructor	classroom with DWAN stations and projector	50	50	200
PO 006: Act as a DRMIS Supervisor				Total PO time:		1195
PO ATT:	1195	PO Admin Time:		PO PT time:		
EO 006.01	Navigate DRMIS SAPGUI	DRMIS Instructor	Classroom w/DRMIS simulator	95	15	
EO 006.02	Perform Maintenance Functions in SAPGUI	DRMIS Instructor	Classroom w/DRMIS simulator	920	50	

EO 006.03	Manage Projects using DRMIS SAPGUI	DRMIS Instructor	Classroom w/DRMIS simulator	45	10	
EO 006.04	Utilize DRMIS SAPGUI as a Maint Mgr	DRMIS Instructor	Classroom w/DRMIS simulator	45	15	

Annex G - Scheduling Resourc

ANNEX H – **ADDITIONAL TRAINING RESOURCES**

In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**)

ANNEX I – **COSTING INFORMATION**

Include information about cost in whichever format is applicable (**Not to be deleted**).

Template Annex I Costing.xls

ANNEX J – **COURSE HISTORY**

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details as to the nature and reasons for amendments are to be provided here.

The Marine Technician Supervisor PO2 RQ is an amalgamation of elements of legacy Marine Engineer QL 6, Electrical Technician QL 6A, and Hull Technician QL 6A. This course is a follow-on from the Marine Technician RQ – LS courses (watch-keeping, mechanical, and electrical technician) and USQ MS courses. This course was developed with input from representatives from each of the three aforementioned legacy MOSIDs in an effort to ensure that the appropriate job elements were captured moving forward.

The Marine Technician Manager PO1 RQ stood from 5-16 Mar 18. Recommendations from the board included the addition of **EO 002.01 Make stability recommendations to HQ1.** These additions were made to the QSP on 26 Mar 18. This addition was to allow the PO2 coming from the Electrical stream a basis of knowledge for Stability.

7 – 11 May 18: Post USQ PO2 Mech/Elec Supr Working Group - Further amendments are being made to the RQPO2 Mar Tech Mgr QSP after completion of the USQ PO2 Mar Tech Mech/Elec QSPWBs. Changes/amendments include addition of PO 006 Coordinate Training and Readiness. This was identified during the PO2 Elec Supr QSPWB and after approval from D Nav P&T, the PO was scalared and subsequent EOs and assessments were designed. See Anx C for further details.

13 Sept 22: Steady State Review - DRMIS Course Mod 2 – DRMIS Supervisor – 119097 removed from qualifications granted, added to course prerequisites. DRMIS Course Mod 3 – DRMIS Maintenance Manager – 119098 added to qualifications granted. PO 001 DRMIS Course Mod 2 changed to DRMIS Course Mod 3. DRMIS Course Mod 2 is already contained within USQ S1 MECH/ELEC, DRMIS Mod 3 PO's already contained in Annex D & G, appears Mod 2 course codes were entered in error and intent was always Mod 3. Supervisor's Safety Course – 300975 (HRMS) removed from qualifications granted this course is no longer supported or offered by Formation Safety & Environment (FSE). Supervisor's Safety Course – 300975 (HRMS) removed from EO 003.02. General Safety Supervisor Course - 303409 added to course prerequisites, all content is available through DLN. PO 002 all EO renumbered after EO 002.01 Make stability recommendations to HQ1 removed. EO 002.01 Make stability recommendations moved to USQ PO2 Mechanical. EO 003.01 Ship's CO, guest speaker changed to MS ENG post HOD. Annex A completed. All PO's references changed to standardized

format. QSPWB Scalar corrected. Annex G corrected, course duration corrected all PO timings corrected. Development notes added.

30 Mar 23: TM Review for Approval – Removed TBC and "Fix this" comments from various parts of QS and TP. Standards statement in EO 003.02 is weak in current state. LOTO is trained at multiple levels and the supervision of the LOTO program on ships is conducted by the Electrical Personnel.

ANNEX K – **ASSESSMENT PLAN**

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
PC 001	PO 001	To be assessed during DRMIS Course Mod 2 – DRMIS Supervisor			
EC 002.02	EO 002.02	To assess the trainee's ability to coordinate section base team training.	Practical EC	SAT on Crit 7/10 overall	Remedial training/counseling as required.
EC 002.03	EO 002.03	To assess the trainee's ability to brief team members on DC considerations.	Practical EC	SAT on Crit 80% overall	Remedial training/counseling as required.
EC 002.04	EO 002.04	To assess the trainee's ability to coordinate a section base team in actions and emergencies.	Practical EC	100%	Remedial training/counseling as required.
PC 002	PO 002	To assess the trainee's ability to act as section base IC.	Practical PC	100%	First failure: retrain, retest Second failure: TRB IAW NPTORD 4500-54
EC 003.01	EO 003.01	To assess the trainee's ability to manage the Captain's Ship's Book (Engineering).	Practical EC	SAT on Crit 60% overall	Remedial training/counseling as required.
EC 003.02	EO 003.02	To assess the trainee's ability to enforce adherence to SEMS.	Written EC	60%	Remedial training/counseling as required.
PC 003	PO 003	To assess the trainee's ability to adhere to environmental health and safety regulations.	Practical PC	SAT on Crit 60% overall	First failure: retrain, retest

					Second failure: TRB IAW NPTORD 4500-54
EC 004.01	EO 004.01	To assess the trainee's ability to coordinate departmental maintenance.	Practical EC	60% overall	Remedial training/counseling as required.
EC 004.02	EO 004.02	To assess the trainee's ability to coordinate marine engineering trials.	Written EC	60%	Remedial training/counseling as required.
PC 004	PO 004	To assess the trainee's ability to coordinate marine systems maintenance.	Practical PC	60%	First failure: retrain, retest Second failure: TRB IAW NPTORD 4500-54
PC 005	PO 005	To be assessed during Naval Technical Communication Module C			
EC 006.02	PO 006	To assess trainee's ability to coordinate personnel employment.	Practical EC	70% of EC checklist items	Remedial training/counseling as required.
PC 006	PO 006	To assess trainee's ability to coordinate training and readiness.	Practical PC	80% of PC checklist	First failure: retrain, retest Second failure: TRB IAW NPTORD 4500-54

ANNEX L – ***TECHNOLOGY ENABLED LEARNING (TEL) PLAN***

- To facilitate section base IC training to minimize the dependence and overloading on coastal DCTFs.
- DCTF trainers are projected to be updated which will mean that one trainer will be available on only one coast at a time. Period 1 year per training facility. Following updates to trainer, there will be a backlog of training.
- Small network which mimics the BDCS onboard HMC ships. Could be linked into the existing IPMS training at NFS.
- L3-Mapps is the current manufacturer for CPF, future ships will have a similar system.
- Basic system to consist of three BDCS screens, networked, with software to mimic shipboard DC including manning components and overlays.
- Could be used to run other courses concurrently (IE: OOD/OOW course)
- Reserve training during summer months could impact availability of DCTF
- Scheduled shutdown periods (2/year 3 weeks) will impact availability

Three COAs

1. BDCS trainer procurement for each coasting NFS
2. Existing BDCS at DCTFs – Limited by existing course demand
3. Onboard ship after hours – limited by ships availability, ships schedule, work periods, TRP & ST(P)/ST(A)

		COA 1 **TEL Delivery**	**COA 2** **Interim Delivery**	**Impact on Trg/Assessment if TEL not implemented** **- Realized benefits** **- E/W Coast Delta**
Mod	**EO**			
	002.03	2x 36”-40” touchscreen monitors to be linked into existing IMPS (MCRT) trainers at NFS(P/A) MSDiv. Software patch required to enable linked operation.	Use of existing MCRT trainers, DCTF trainers, or access to HMC ships.	Trainees would not be able to develop a full competency of managing multiple inputs. A key component of Mar Tech supervisor. The more realistic the training, the more efficient the trainee will be in a real-world scenario. Training taking place at NFS MSDiv(s) will save time and money by not having to use the coastal DCTFs (transportation, food services, limited resources, time).